At the start

1. Safety contract collection (did you fill in “I printed name”)
2. Seating chart
3. Park phones in your number spot
4. Who am I? (fill out this week)

Get to Know your Neighbors!

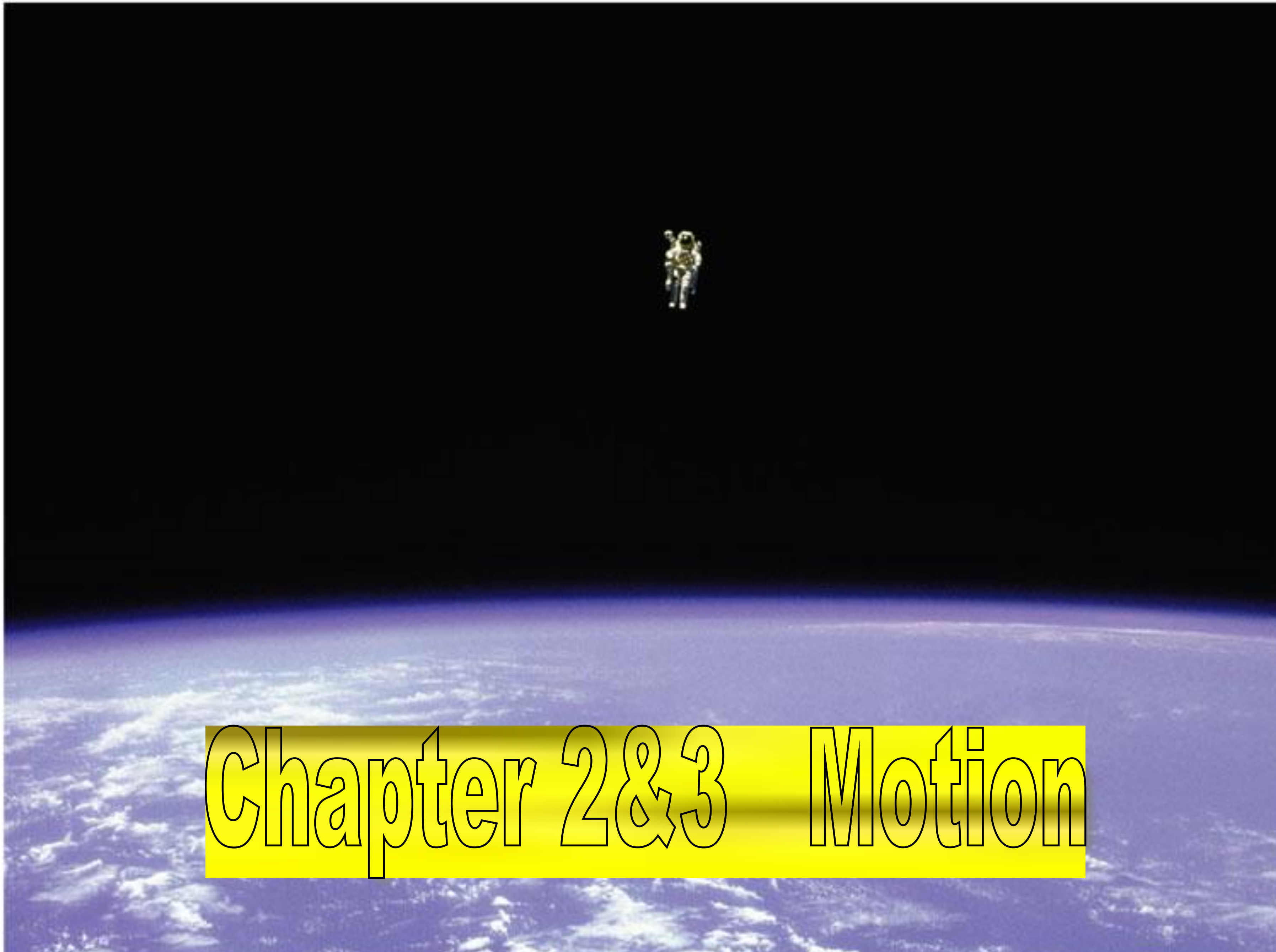
LET'S PLAY!! ***Maroooned***

Introduce yourselves to your team of 4.

Then, you will be given five minutes to find five items from your book bag, pockets, pencil pouch, lunch, or purse (NO CELLPHONE COVERAGE ON THE ISLAND) to help your group survive.

Then, you will be given another 5 minutes to discuss with your group to figure out which items are the best to include. Also discuss any experience you've had that might help the group.

A member of the group will be asked to explain in front of the class how each object is essential to your survival.



Chapter 2&3 Motion

8/18 Agenda for the Day

Learning Target: I can use words/symbols to scientifically describe motion.

Materials:

- Note paper
- Pencil/pens

Motion (Do not write this!)

- **"Of all the intellectual hurdles which the human mind has confronted and has overcome in the last fifteen hundred years the one which seems to me to have been the most amazing in character and the most stupendous in the scope of its consequences is the one relating to the problem of motion."**

Sir Herbert Butterfield

Motion

Learning Target

I can use words to scientifically describe motion. This means that I can accurately, precisely and completely describe motion in words.

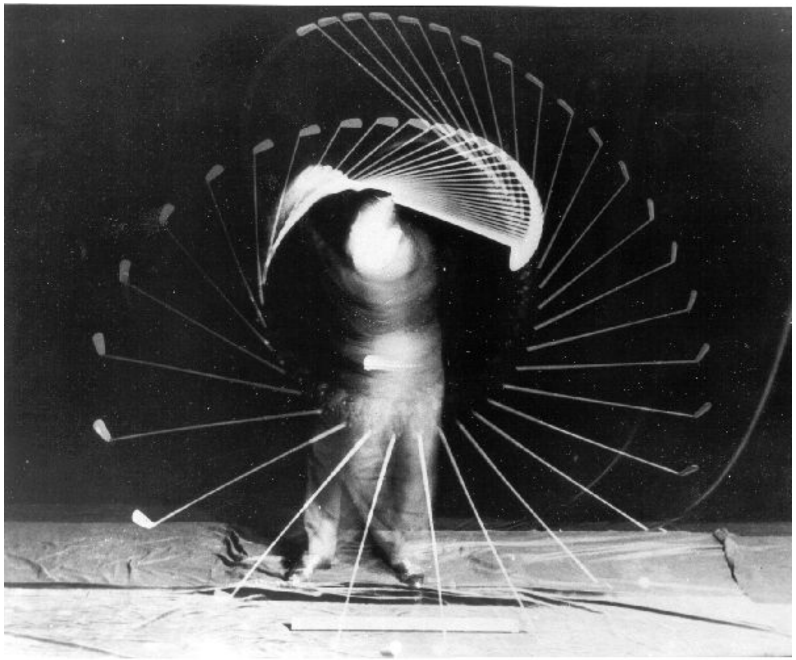
Electrons vs. Hydrocarbon

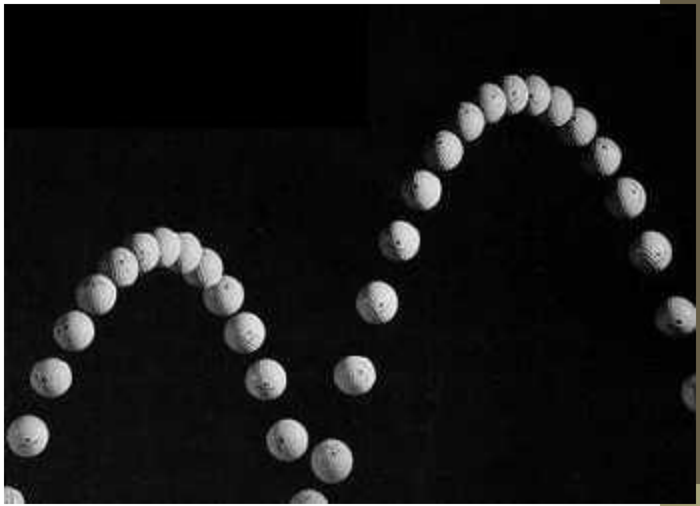
[The race electric vs petrol]

(start at 3:50ish)

1. Watch the race
2. Retell the story of the race using 2 different styles. Consider pictures, symbols, graphs, math, arrows, diagrams, etc. Use what seems to be the most efficient method…..

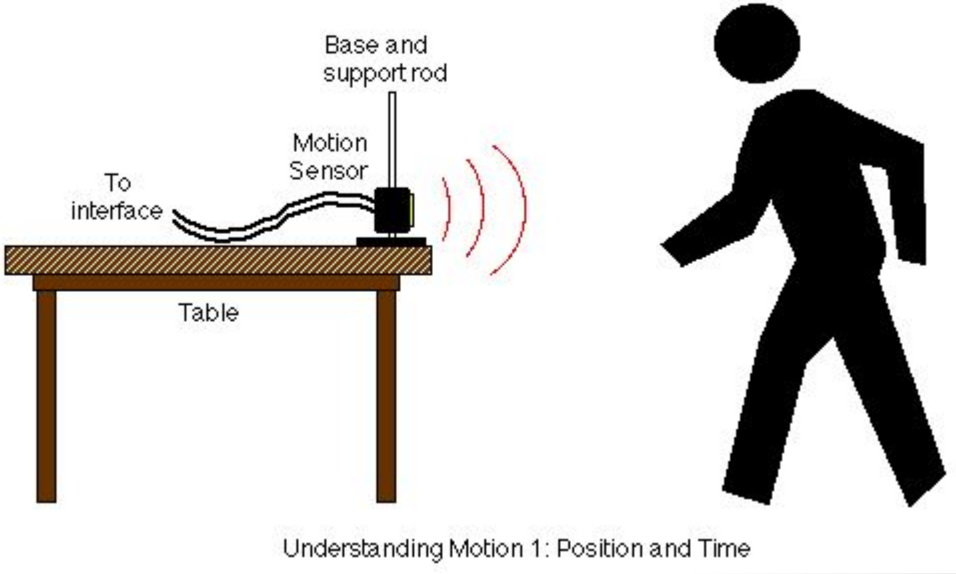
2.1 Picturing Motion





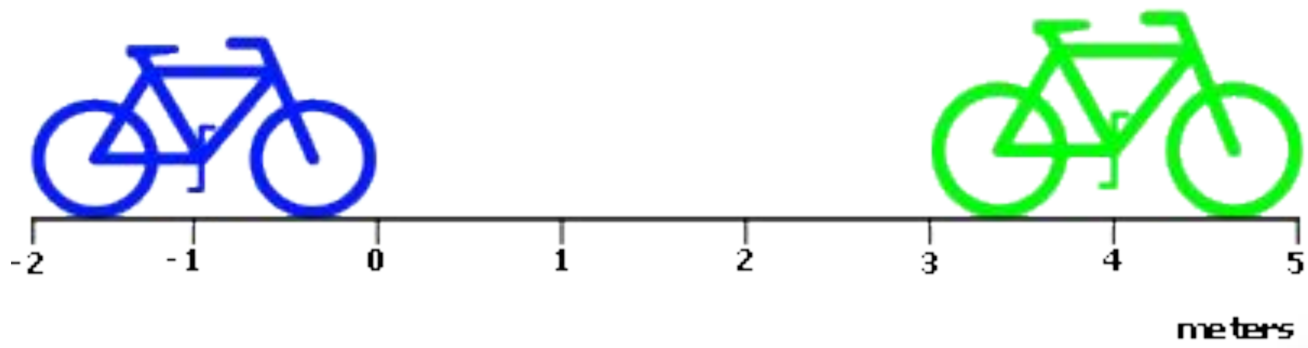
Section 2-2:Where and when

- An object is in motion if its distance from another object is changing.
- An object is in motion if it changes position relative to a reference point.



Reference point or coordinate system

- Reference point is a place or object used for comparison
- A coordinate system tells you the location of the zero point
- Position can be indicated as a distance in the positive or negative direction from the reference point or separation from the origin.



Fundamentals: Measurement

With your desk partner, complete the Measurement activity.

Use the ruler, meterstick or scale as appropriate.

Answer check

New Mile record!

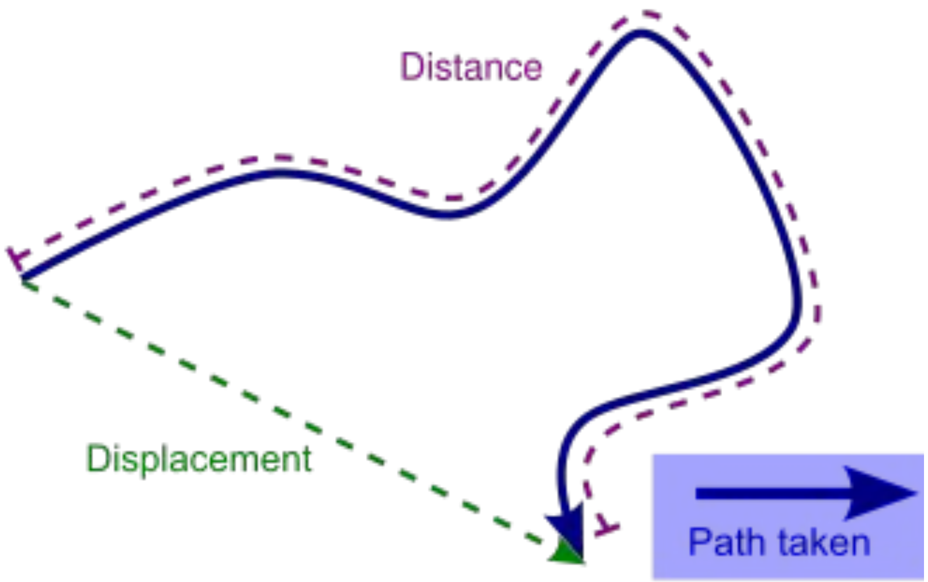
Where? How far?

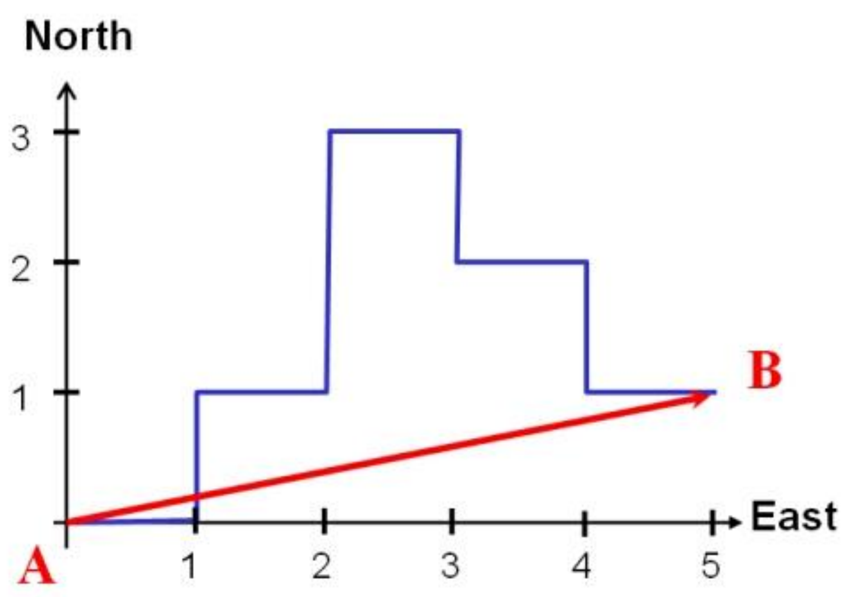
Position = a location of object relative to reference point or origin

Distance = total length of the **actual path** between to points

Displacement = length of a straight line between the starting and ending points (includes the direction of travel)

Distance versus displacement

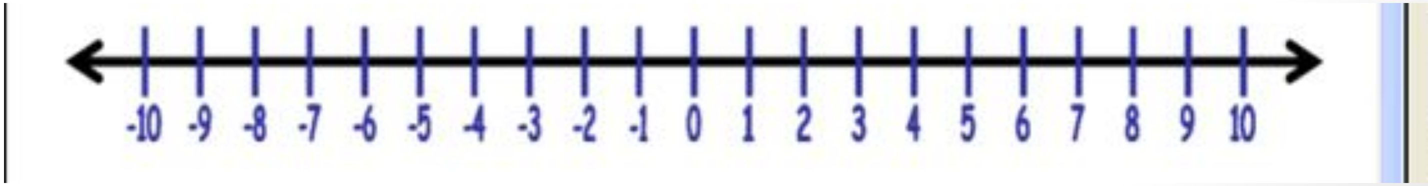




Is distance dependent or independent of path? Displacement?

Finding position, distance and displacement

- A person starts on the 2 m mark and walks to the 8 m mark and then runs back to the -2 m mark.
- Find the total distance AND the total displacement.



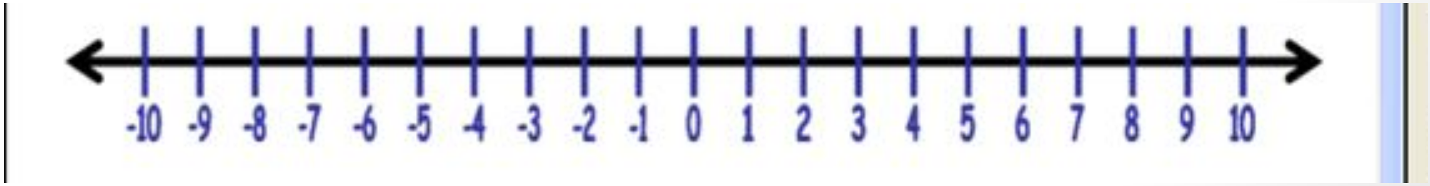
Finding position, distance and displacement

- A person starts on the 2 m mark and walks to the 8 m mark and then runs back to the -2 m mark.

- Find the total distance AND the total displacement.

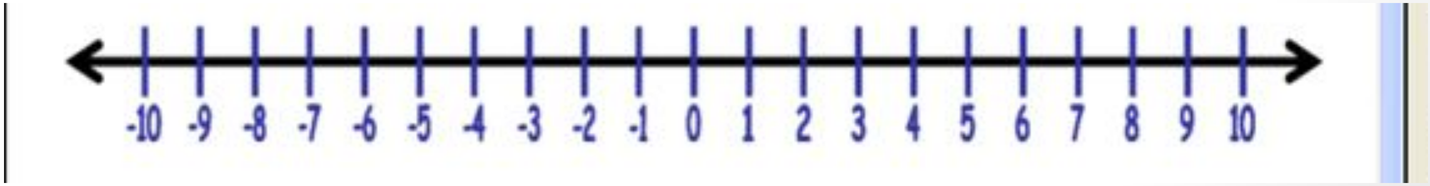
(Distance = 6 m + 10 m = **16 meters**

Displacement = from +2 to -2 on the number line would be **4 meters backward (vector)**



Finding position, distance and displacement

- A person starts on the 2 m mark and walks to the 8 m mark and then runs back to the -2 m mark.
- Find initial position, position at end of walk, position at end of run.
- Find distance traveled during walk, run and total distance traveled.
- Find the displacement for the walk, run and the total displacement.



A person starts on the 2 m mark and walks to the 8 m mark and then runs back to the -2 m mark.

- **Position**
  - initial position = 2 m,
  - position at end of walk = 8 m,
  - position at end of run = -2 m.
- **Distance**
  - Walk = 6 m,
  - Run = 10 m
  - Total distance traveled = 16 m.
- **Displacement**
  - Walk = 6 m forward (+ 6m)
  - Run = 10 m back (-10 m)
  - Total displacement =4 m back (- 4 m)

Vector and scalar quantities

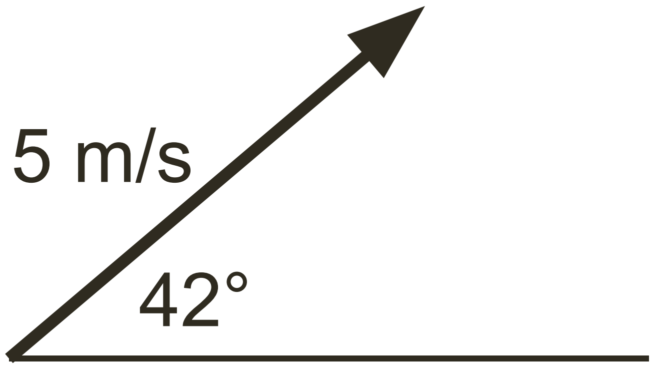
- Vector quantity shows both magnitude (length or size) and direction. eg **55 miles north**
- Scalar quantity only includes length or size (magnitude) **(50 miles)**

- Vector or scalar quantity?
  position
  distance
  displacement

Vectors

Vectors are represented with arrows

- The length of the arrow represents the magnitude (how far, how fast, how strong, etc, depending on the type of vector).
- The arrow points in the directions of the force, motion, displacement, etc. It is often specified by an angle.



Some Physics Quantities

Vector - quantity with both magnitude (size) and direction

Scalar - quantity with magnitude only

<u>Vectors</u>:

- Displacement
- Velocity
- Acceleration
- Momentum
- Force

<u>Scalars:</u>

- Distance
- Speed
- Time
- Mass
- Energy

When?

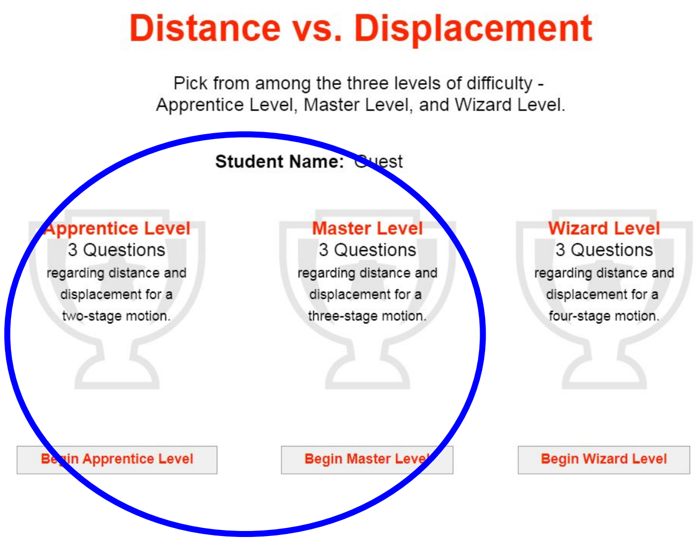
Time interval = the difference between two times Δt

Time = the clock reading, "t"

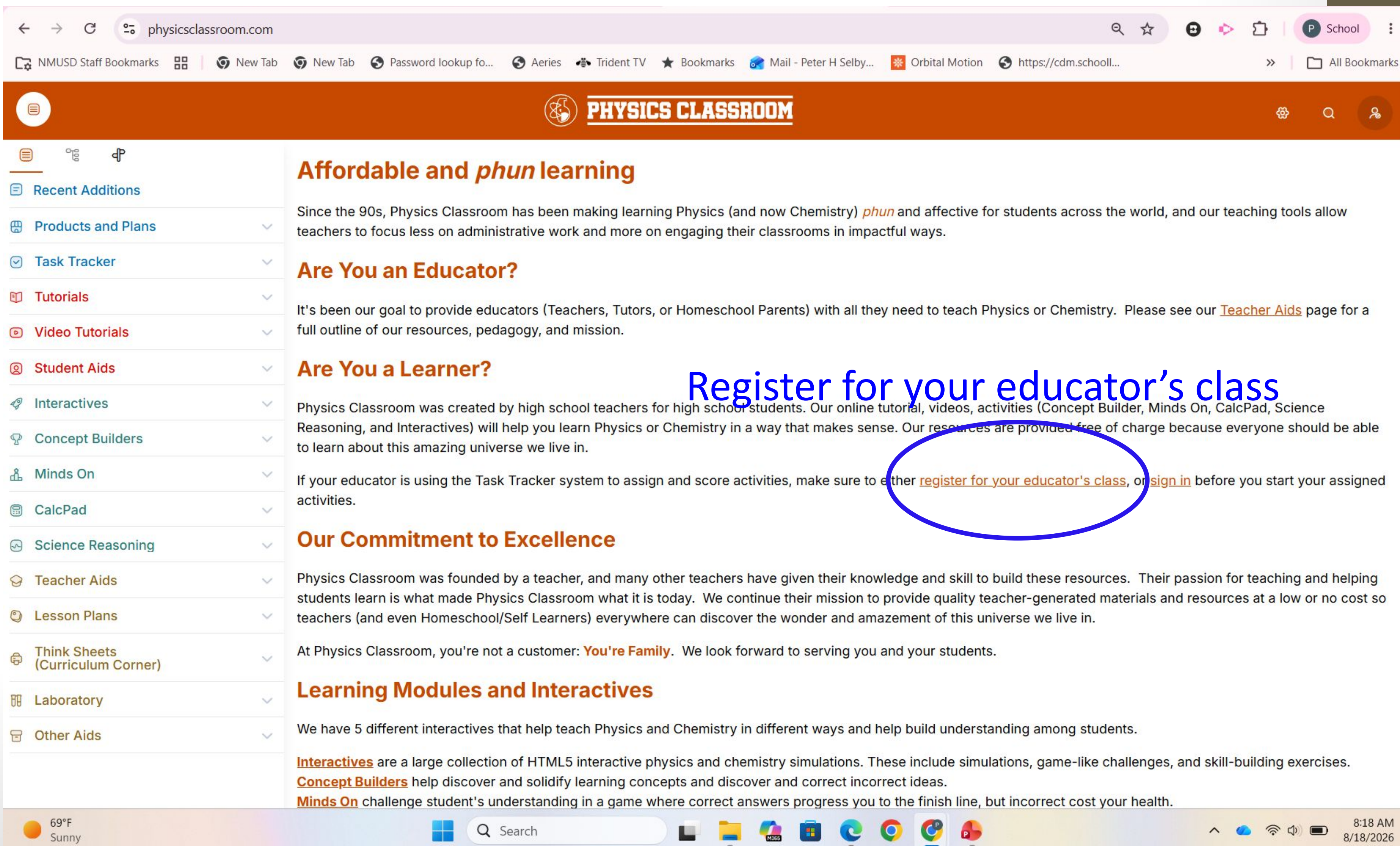
(a clock reading has no duration)

Distance vs. Displacement

- www.physicsclassroom.com
- go to Task Tracker
- “Distance vs Displacement”
- Click “Launch”
- Do Apprentice & Master levels



www.physicsclassroom.com



Register for your educator’s class

- Fill in your info (including an email address)
- Course Codes (use your period)

  Period 1 = a37afee

  Period 2 =fa8a7cc

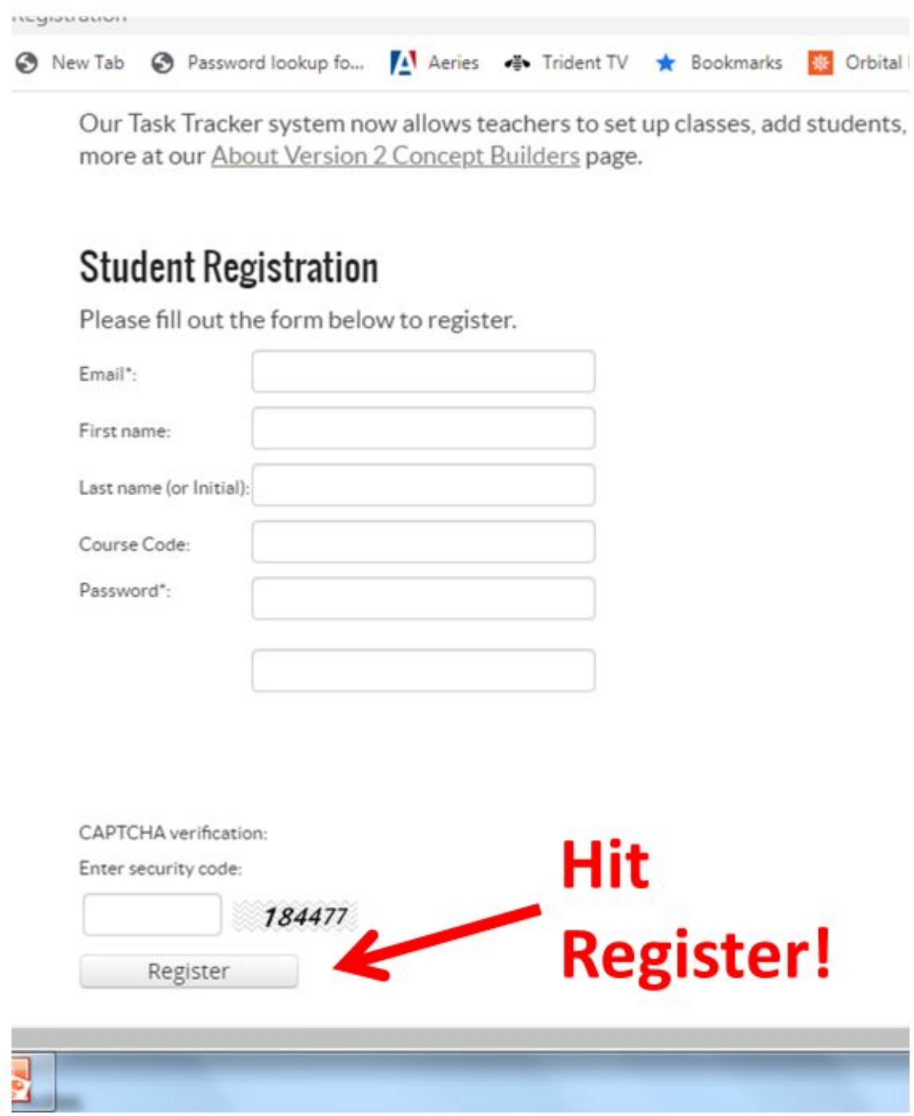
  Period 3 = ecf011f

  Period 4 = 48c1b2e

  P 5 AP = 7a05d22

  Period 6 = 1b326aa

- Password (can use your student ID # from CdMHS)
- Hit "Register"



End of Day 1 (HW)

1. Finish Concept Builder
2. Submit your “Who am I” slide
3. Lab Safety form (sign and return to Mr.S)

8/20 - 8/21 Agenda

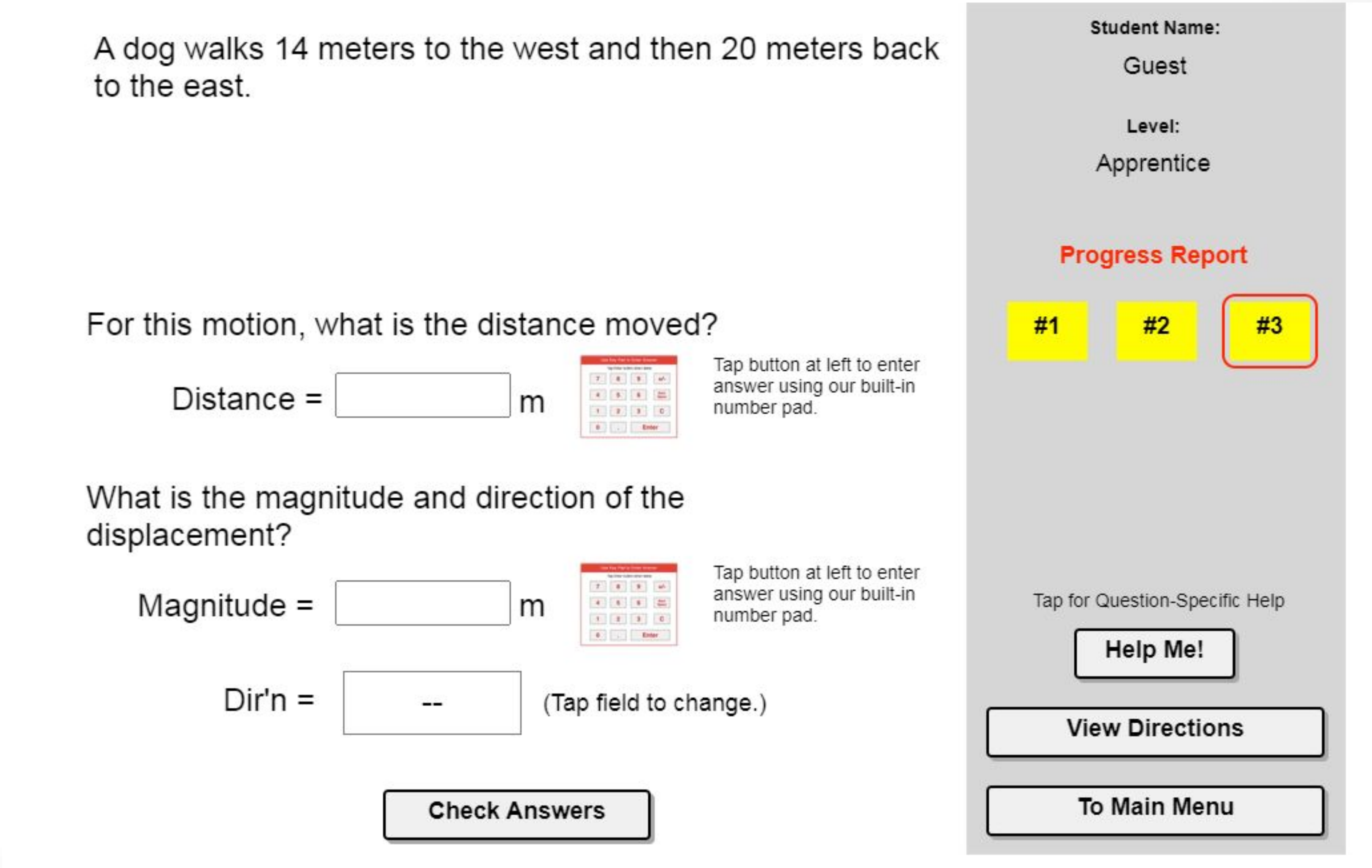
1. Park phones!
2. Turn in Lab Safety forms
3. Warm-up
4. Practice w/ describing motion
5. Notes
6. Lab Activity (Tumble Buggy)

Partner check in…

What would be your "walk up" song?

When you walk up to the big stage….

Review from last class



Answer check

A dog walks 14 meters to the west and then 20 meters back to the east.

For this motion, what is the distance moved?

Distance = 34 m

Tap button at left to enter answer using our built-in number pad.

What is the magnitude and direction of the displacement?

Magnitude = 6 m

Tap button at left to enter answer using our built-in number pad.

Dir'n = East (Tap field to change.)

Check Answers

Terms Practice wkst

Helpful symbols:

x_0 = initial position
x = final position
$\Delta x = (x - x_0)$ = displacement

t = time interval

Learning Target (Motion cont.)

- I will be able to measure and calculate the rate of motion of an object (speed or velocity).

(get note paper handy)

Speed and velocity: How fast?

- Speed = how fast is the object moving (# only, no direction)
- Average velocity = displacement over a time interval (longer interval) in a direction
- Instantaneous velocity = displacement during an instant (tiny interval of time) in a direction

Does a speedometer measure average speed, instantaneous speed or velocity?

Speed vs. Velocity

- Speed is a scalar (how fast something is moving regardless of its direction).
  Ex: v = 20 mph
- Speed is the magnitude of velocity.
- Velocity is a combination of speed and direction. Ex: $\mathbf{v}$ = 20 mph at 15° south of west
- The symbol for speed is v.
- The symbol for velocity is type written in bold: $\mathbf{v}$ or hand written with an arrow: $\vec{v}$

Speed vs. Average Velocity

- You decide to drive to Los Angeles. Early in the trip, you glance down at the speedometer and see 70 mph. Several minutes later, you encounter traffic and notice that you are going 30 mph. For the entire trip you never go faster or slower than these values. What is a possible average velocity for the trip to LA?

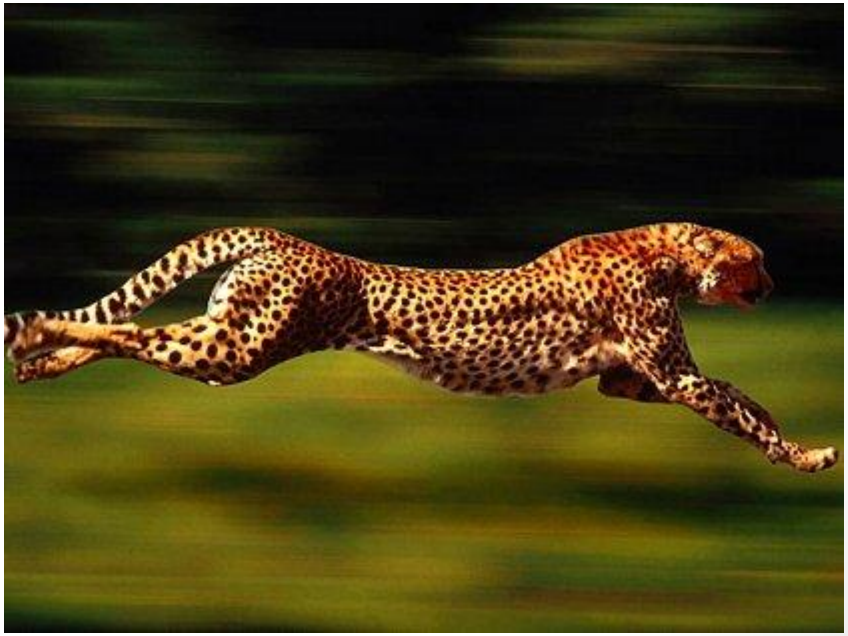
Velocity

how fast #

- Changes in velocity may be due to changes in speed, changes in direction or both.
  - Vector or

scalar quantity?

If the leopard veers left but continues at the same speed does his velocity change?



Calculating average velocity

Average velocity = **<u>displacement</u>**

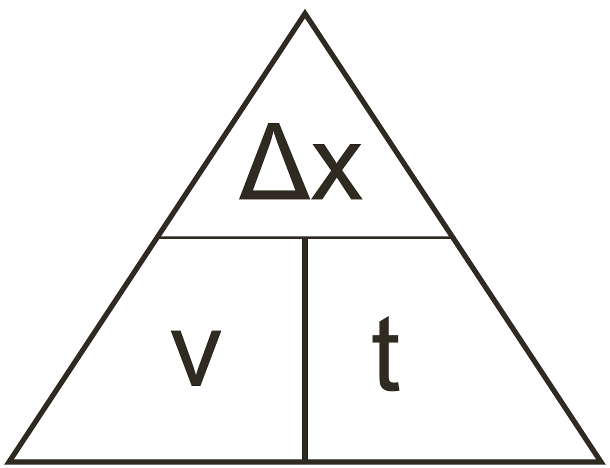
time interval

How could you express this as an equation? (v = velocity, $(x - x_0)$ = displacement , t = time)

Units (meters/second, miles/hr)

- $V = \Delta x/t$
  - When velocity is constant;

    average velocity = velocity
  - When start at zero mark and move in one direction: distance = displacement
  - When initial clock reading is zero; $\Delta t = t$



Units

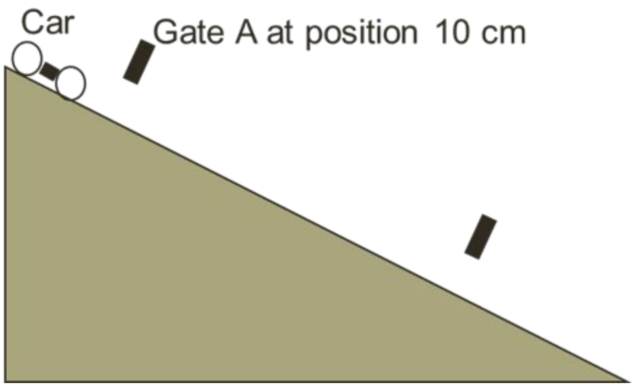
Units are not the same as quantities!

Quantity . . . Unit (symbol)

- Displacement & Distance . . . meter (m)
- Time . . . second (s)
- Velocity & Speed . . . (m/s)
- Acceleration . . . (m/s^2)
- Mass . . . kilogram (kg)
- Force . . .Newton (N)

Activities (Car & Ramp / Tumble Buggy)

Part A - Car & Ramp



Part B - Tumble Buggy

-Find the velocity of Buggy (using stopwatch, meter stick, tape, etc.).

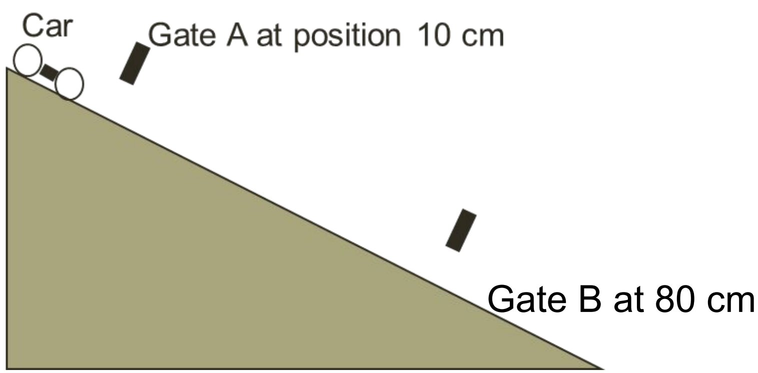
-Use floor space anywhere inside the room. (Floor tiles are 1 foot square).

-Record data, show work (**Units too!**)

Example data

Width of wing on car = 1 cm

(popsicle stick)



Time in Gate A = _0.02___

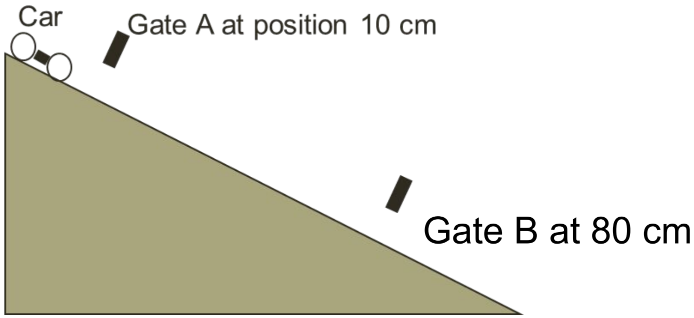
Time in Gate B = _0.01___

Time from A-B = __0.4__

Time AB (both lights)	Distance AB (cm ok)	Popsicle Stick width (cm)	Time in A (light A only)	Time in B (light B only)	Velocity A = d/tA	Velocity B = d/tB

Width of wing on car = 1 cm

(popsicle stick)



Time in Gate A = _0.02___

Time in Gate B = _0.01___

Time from A-B = __0.4__

Time AB (both lights)	Distance AB (cm ok)	Side wing width (cm)	Time in A (light A only)	Time in B (light B only)	Velocity A = d/tA	Velocity B = d/tB
0.02 Or 0.4?	10, 70 or 80 cm?	1	0.02 or 0.01?		1/0.02 10/0.02 Or 10/0.4?	

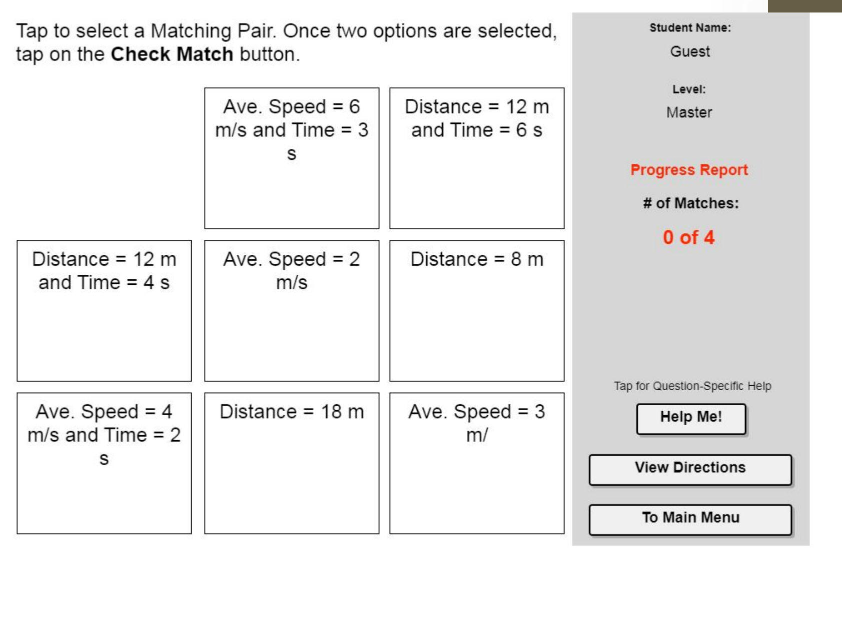
Apply Motion Terms

- Watch the America’s Cup race from Bermuda between USA and New Zealand.
- Find examples of our motion terms in the broadcast.
- Jot down as many examples as you can find

- (start at 10 minute mark)
- https://www.youtube.com/watch?v=iJEN4yktEHc&t=385s

Check for Understanding

- www.physicsclassroom.com
- Task tracker
- "Speed-distance-time"
- Launch
- Matching Pairs
- (complete Apprentice & Master)



HW

- Finish Concept Builders “Speed - Distance - Time” & “Distance Vs. Displacement” before Monday….

- Who am I?

8/25-26

Agenda

- Park phones
- Get out your Avg Vel HW wkst on desk
- 4 colored pencils & a ruler

Private citizen tickets to space are now available. If you had the option, would you go?
(free) Why or why not?
Talk it over

Review from last class

- Example: An Amazon delivery truck drove 100 meters North in 20 seconds, 200 meters North in 10 seconds and 300 meters North in 30 seconds. What is the average velocity of the truck?

(Hint: Avg Vel = $\frac{Total\ Displacement}{Total\ Time}$)

Solution:

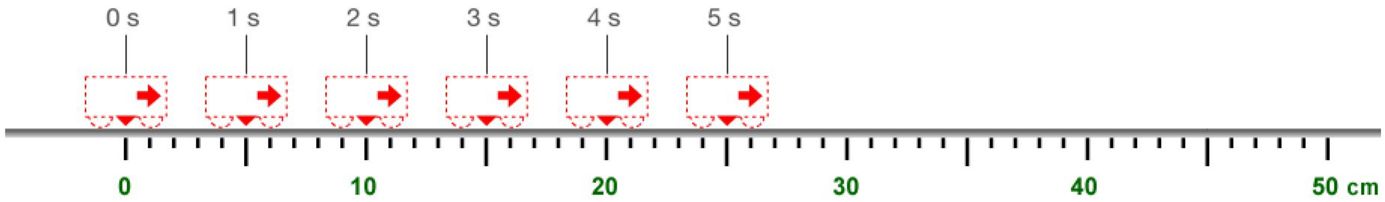
- Example: An Amazon delivery truck drove 100 meters North in 20 seconds, 200 meters North in 10 seconds and 300 meters North in 30 seconds. What is the average velocity of the truck?

(Hint: Avg Vel = $\frac{Total\ Displacement}{Total\ Time}$)

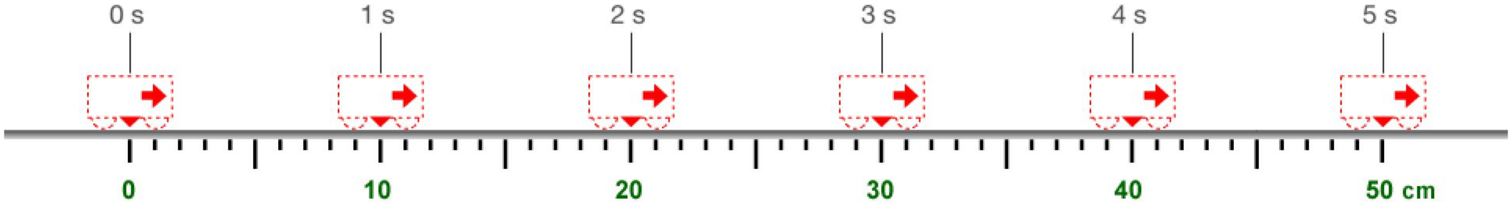
Learning Target

- I will be able to describe acceleration in words and motion diagrams. I will recognize the difference between uniform and non-uniform motion.

Find velocity from Dot Diagram

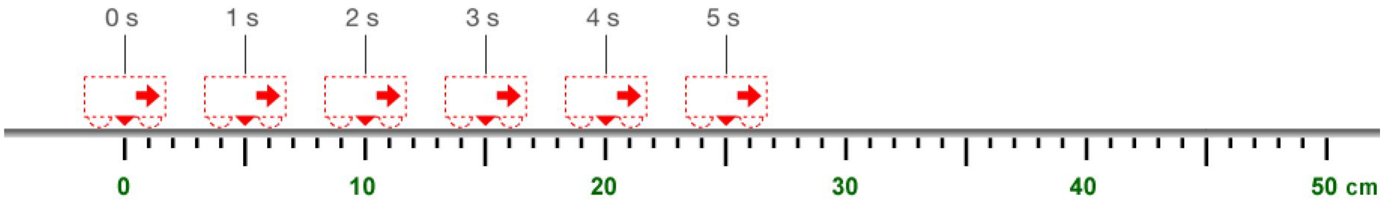


What is the velocity of the car above? Use evidence shown in diagram…

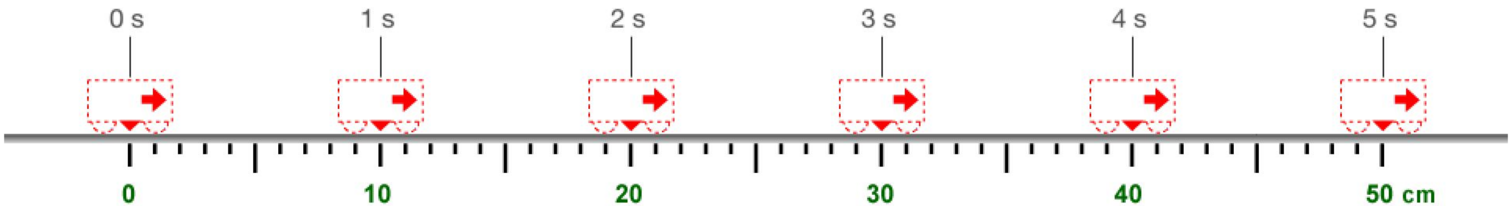


What is the velocity of the car above? Use evidence shown in diagram…

Find velocity from Dot Diagram

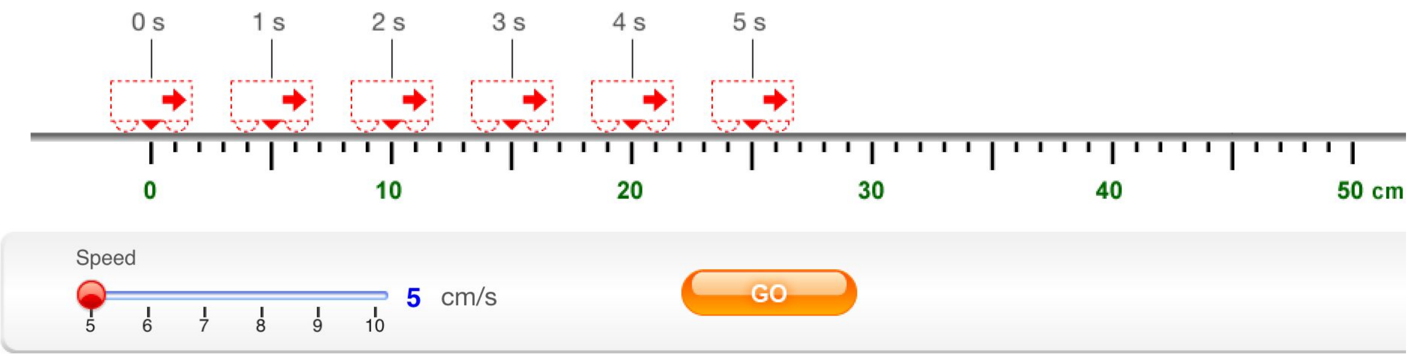


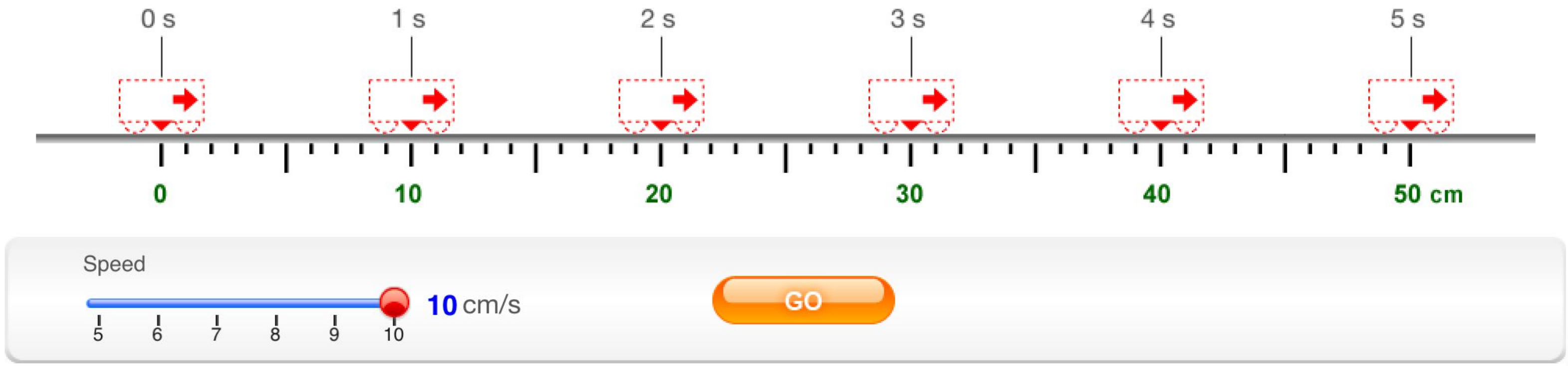
Velocity = 5 (magnitude) cm/s (units) rightward (direction)
V = 5 cm/s right



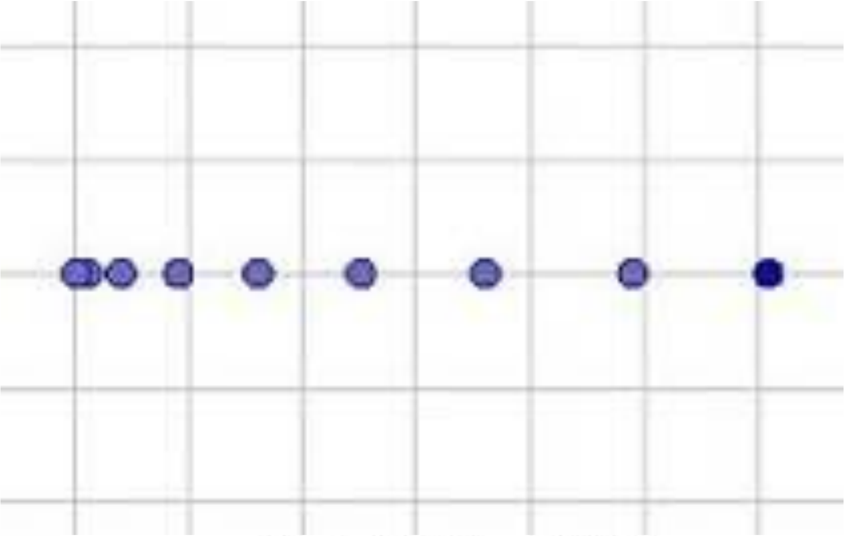
Velocity = 10 cm/s rightward (East, +, etc.)

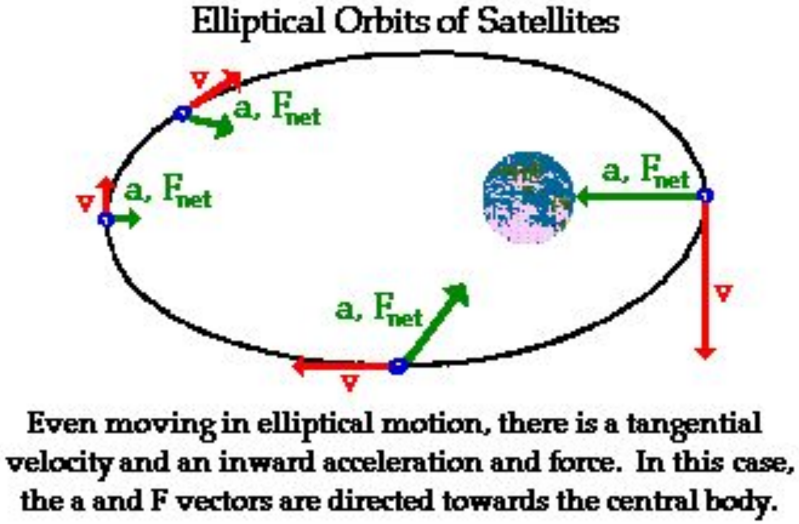
Uniform motion

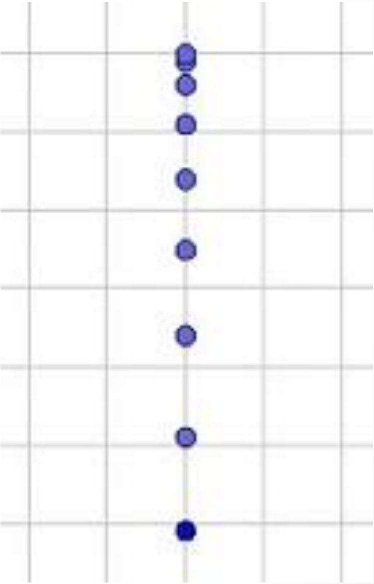




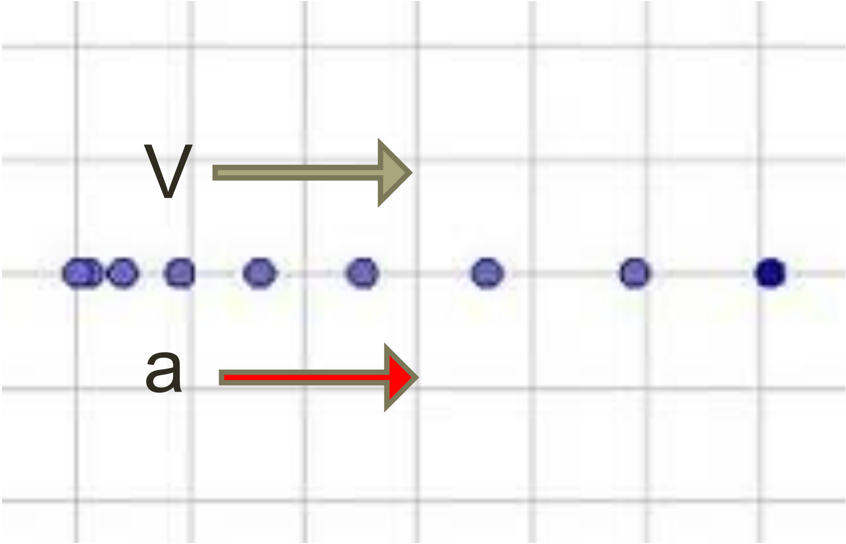
Non Uniform Motion







Example: Acceleration right

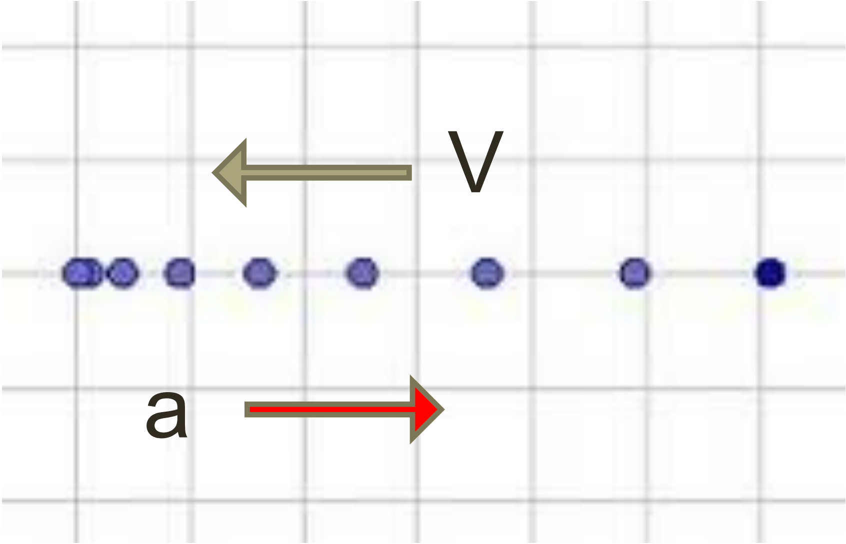


Possible interpretation #1: (in simple terms)

Going faster and faster to the right

Velocity right, Acceleration right

Example: Acceleration right



Possible interpretations #2: (in simple terms)

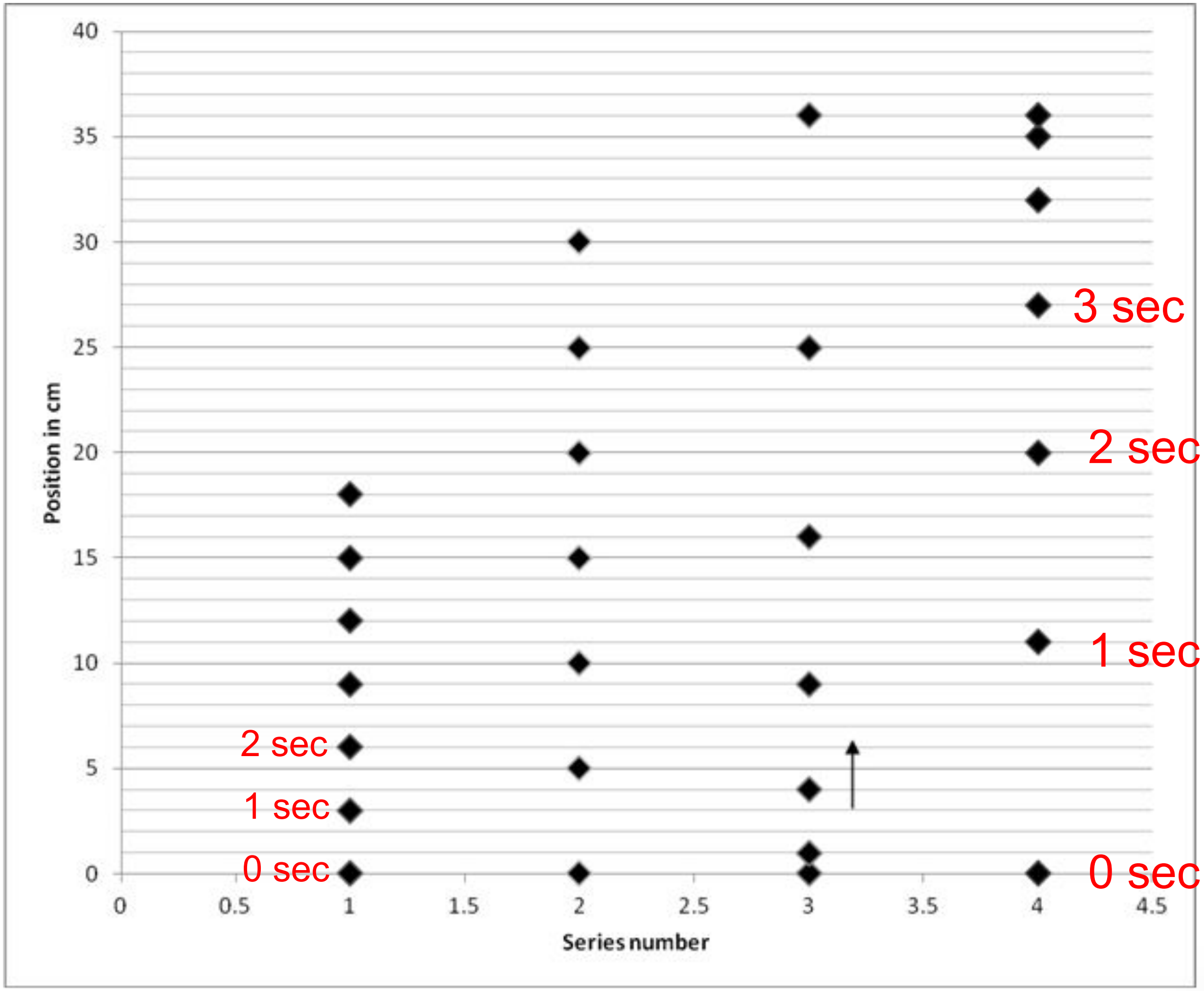
Slowing down while moving left

(Velocity left, Accel right)

Motion Terms

- What is uniform motion? An object moving in a straight line with an unchanging velocity.
- What is non-uniform motion? An moving with changing velocity. Speeding up, slowing down, turning.
- How does a motion diagram demonstrate non-uniform motion? (for example, a person going faster and faster)

Analysis of Motion (Strobe Photo packet)



Take a break

Dance to the Graph…

Click on the link sent to you in Schoology Message (Pace Tracer 1)

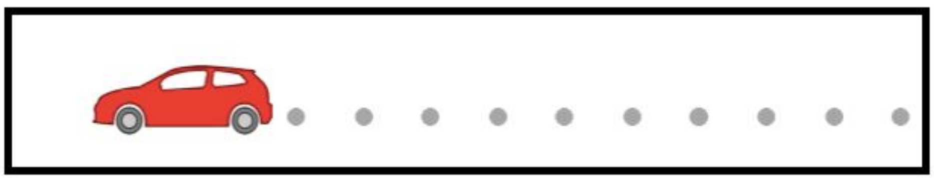
“Launch Interactive” (& allow camera)

Use a QR code to pick a dance, then move as the graph tells you. The closer your motion to the graph, the higher the score.

Show Mr.S an 80% or higher score after you finish a dance…..

Does the Description match the picture?

View the animation below and find a verbal description that matches the motion.



Tap on the Verbal Description to toggle to a different one.

Verbal Description # 1

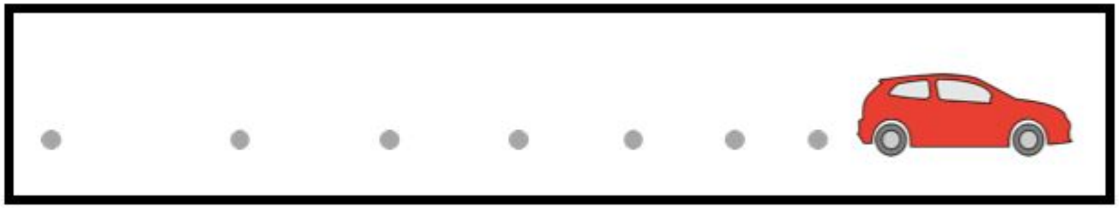
The car moves with a rightward velocity and a rightward acceleration.

When you're convinced that the verbal description above describes the animated motion, tap on the **Check Answer** button.

Check Answer

Does the Description match the picture?

View the animation below and find a verbal description that matches the motion.



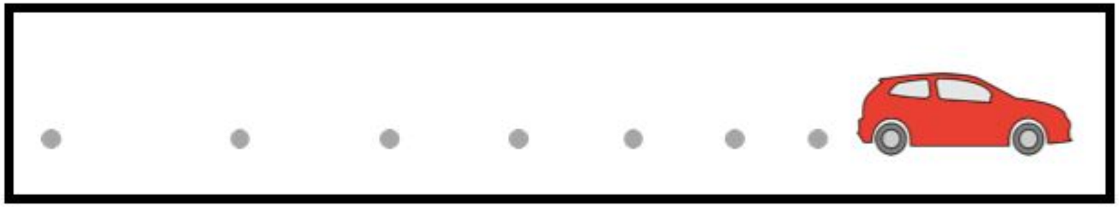
Tap on the Verbal Description to toggle to a different one.

Verbal Description # 1

The car moves with a rightward velocity and a rightward acceleration.

Answer

View the animation below and find a verbal description that matches the motion.



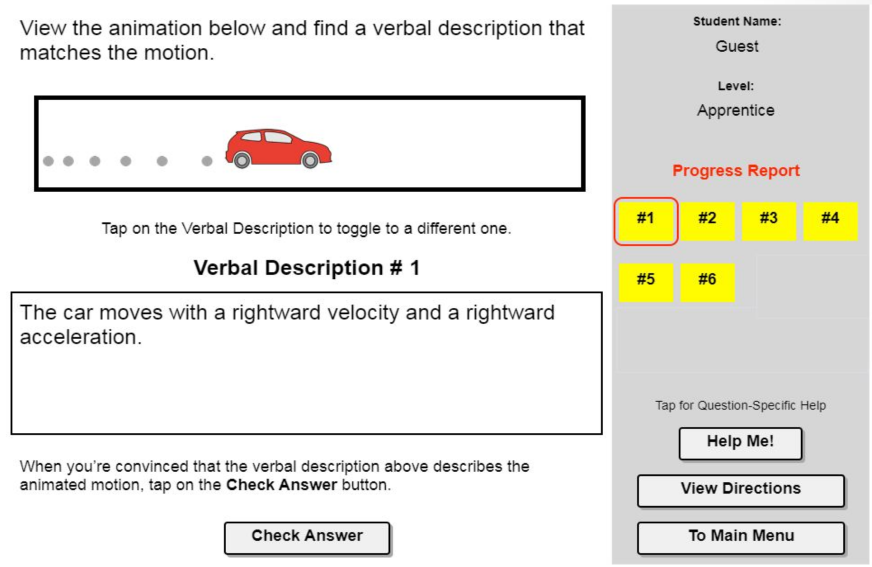
Tap on the Verbal Description to toggle to a different one.

Verbal Description # 1

The car moves with a rightward velocity and a leftward acceleration (the accel is “against” the direction of motion)

Check for Understanding

- Physicsclassroom.com
- Task Tracker
- "Name that Motion"
- Launch
- Do Apprentice and Master levels



Start this now, continue as HW if necessary

End of Day 3

8/27 - 8/28

Park phones

Notes (get out your Analysis of Motion packet)

Practice w motion graphs (whiteboard & marker at your table)

More dancing!

Analysis of Motion

Have the back page visible as we try some practice scenarios

Add some new graphs to the back page (position backward)

Link to Notes page

Scenario 1

Two cross country runners start together. One runner runs forward at a constant velocity for the whole race. The second runner starts slowly and then gradually goes faster and faster. They both finish the race side by side.

Sketch a position graph of the event

Scenario 2

Two football teams get ready for kickoff. Focus on one player from each team. The players run at constant speed toward each other from opposite ends of the field. They collide near mid field.

Sketch a position vs time graph for the event

Scenario 3

You drive slowly to school at constant speed. At school you park for a few minutes (listening to KPOP sound track). Then you realize that you left your presentation poster board at home. You drive <u>faster (constant speed)</u> back home to pick it up.

Scenario 4 (F1 the movie)

Joshua and Sonny are both moving at same fast constant speed toward a turn on a rain soaked track. Sonny slows down to make the turn, while Joshua loses traction and slides off the track (no change in speed).   Consider both cars to be moving forward

Graphing notes

Info from Position vs Time graphs

Graph Practice Concept Builders

www.physicsclassroom.com

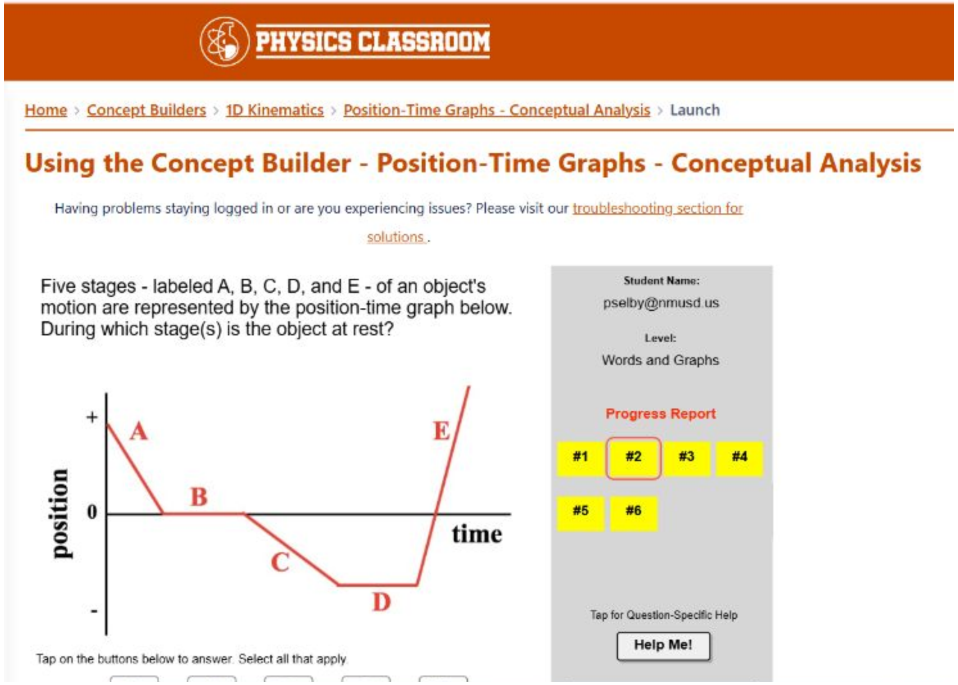
Go to *Task Tracker*

Try the following for Graph practice

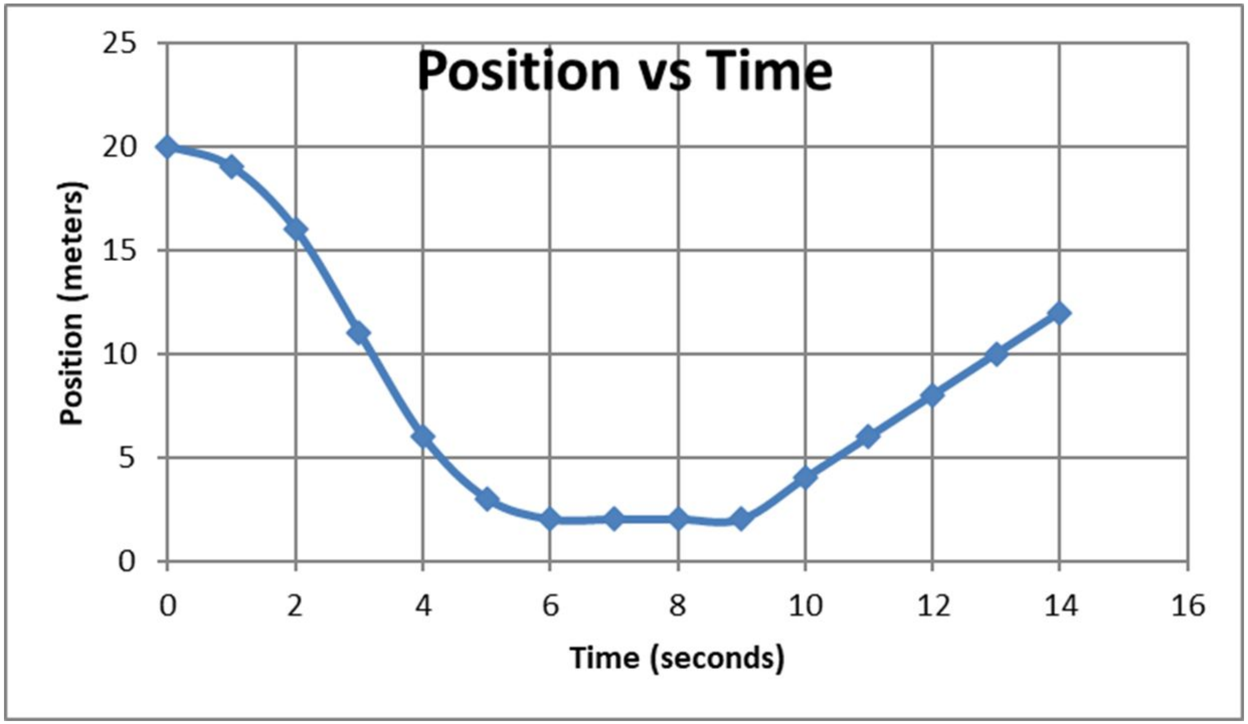
“Position vs Time Conceptual”

(Apprentice & Master)

“Position vs Time Numerical” (Apprentice & Master)



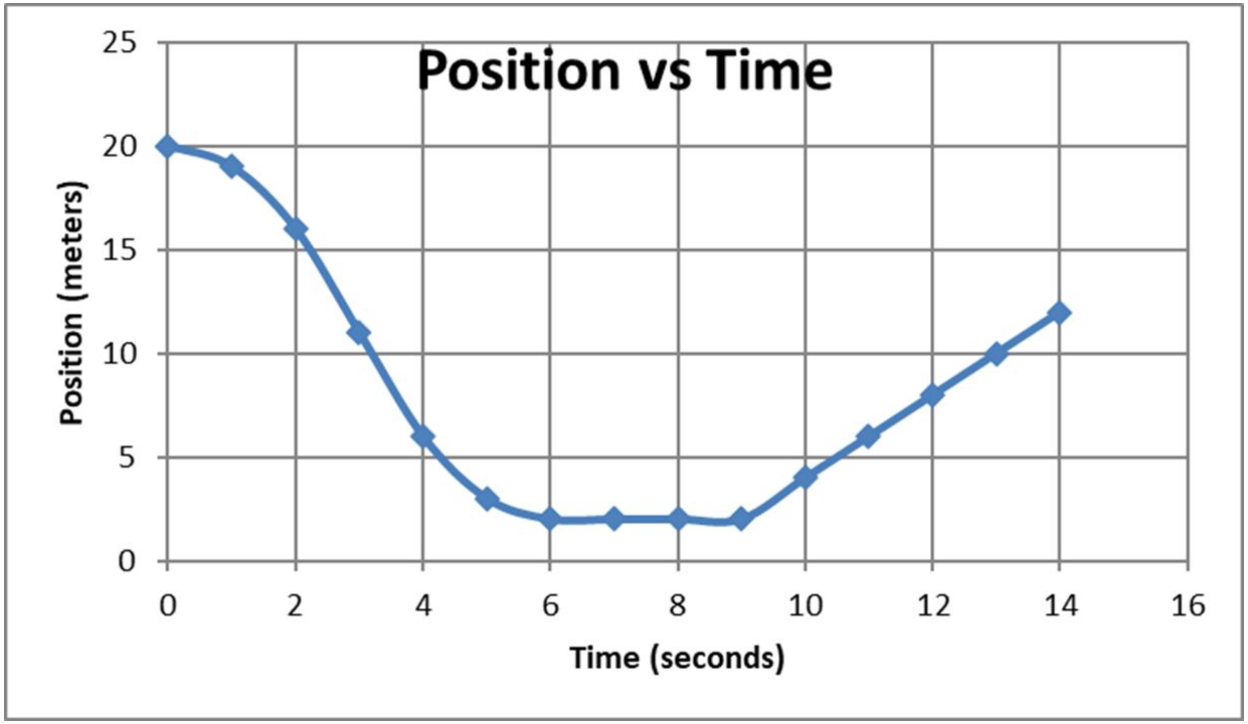
Describe motion in each time interval (tell the story:) (positive is forward)



- Describe motion: 0-3 sec?
- 3-6 sec?
- 6-9 sec?
- 9-14 sec?

Find V (t = 9-14sec)

Describe motion in each time interval (tell the story:)



- Describe motion:
0-3 sec? Faster bwd

- 3-6 sec? Slowing bwd
- 6-9 sec? At rest
- 9-14 sec? Con Vel fwd

Find V (t = 9-14sec)

$V = (x-x_0)/t = (12-2)/5 = 2$ m/s

Velocity vs Time Notes

Velocity Dance Graph

Get the link from Schoology Messages

(don’t need to log in, guest ok)

Launch

Allow camera

(Show me at least 2 gold cups!)

HW:

Finish the Concept Builders….

1. Position vs Time Conceptual
   (Apprentice & Master)

2. Position vs Time Numerical
   (Apprentice & Master)

3. Velocity vs Time (Apprentice only)

8/29

Park phones

Motion graph warmup

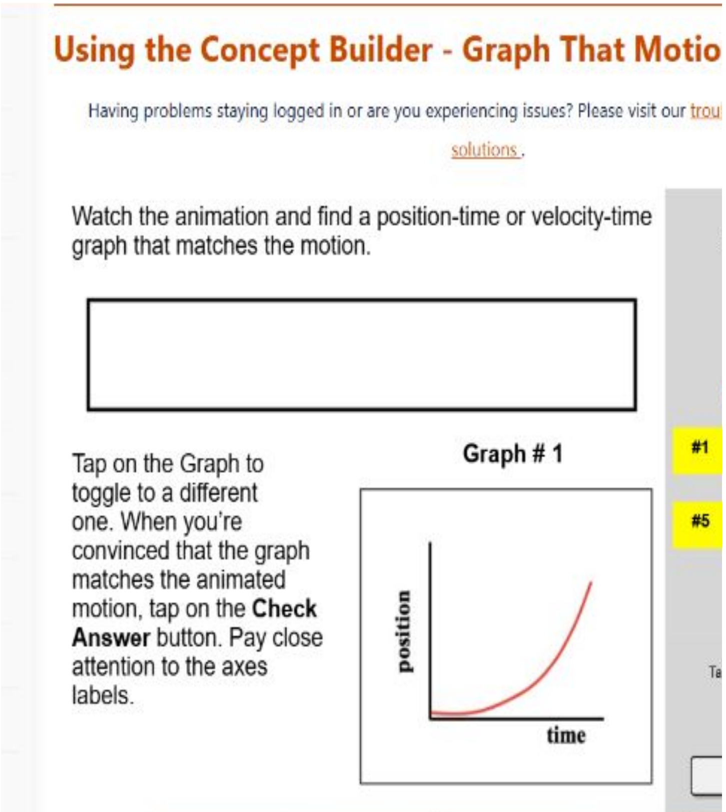
Position Graph Checkup (10 questions)

Graph Practice Concept Builders

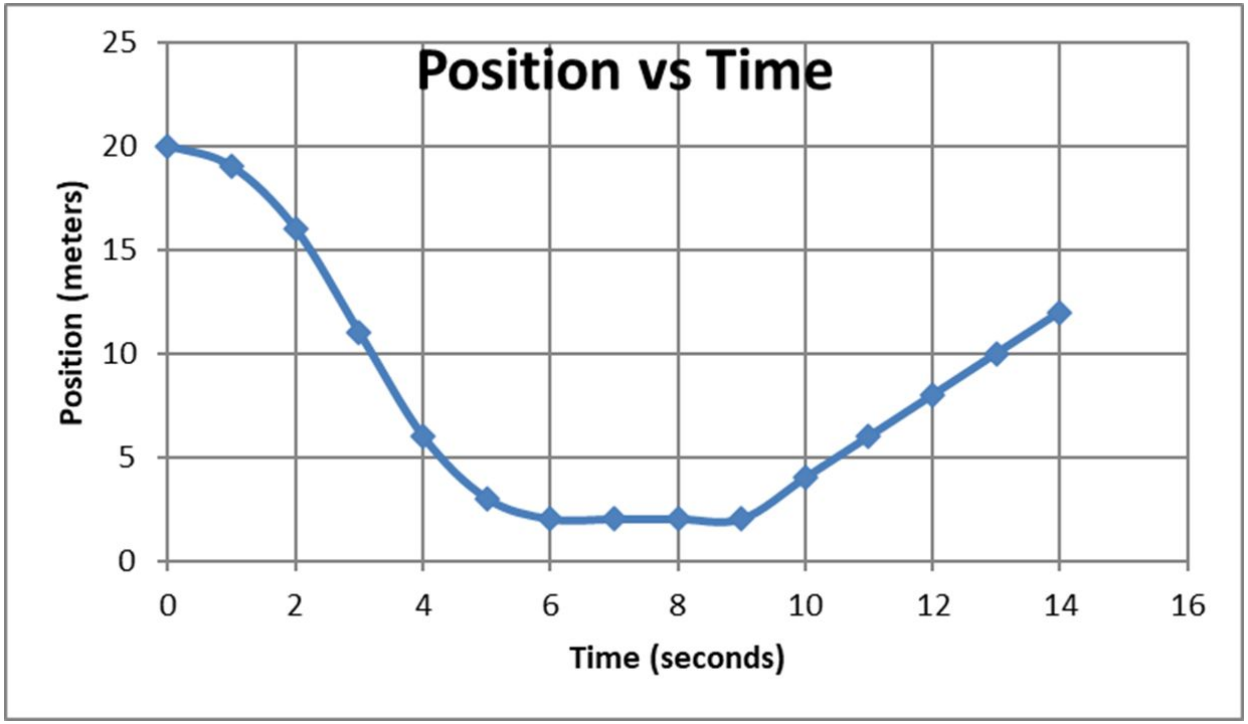
www.physicsclassroom.com

Warm up

“Graph that Motion” (Apprentice)



Describe motion in each time interval (tell the story:) (positive is forward)



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Find V (t = 9-14sec)

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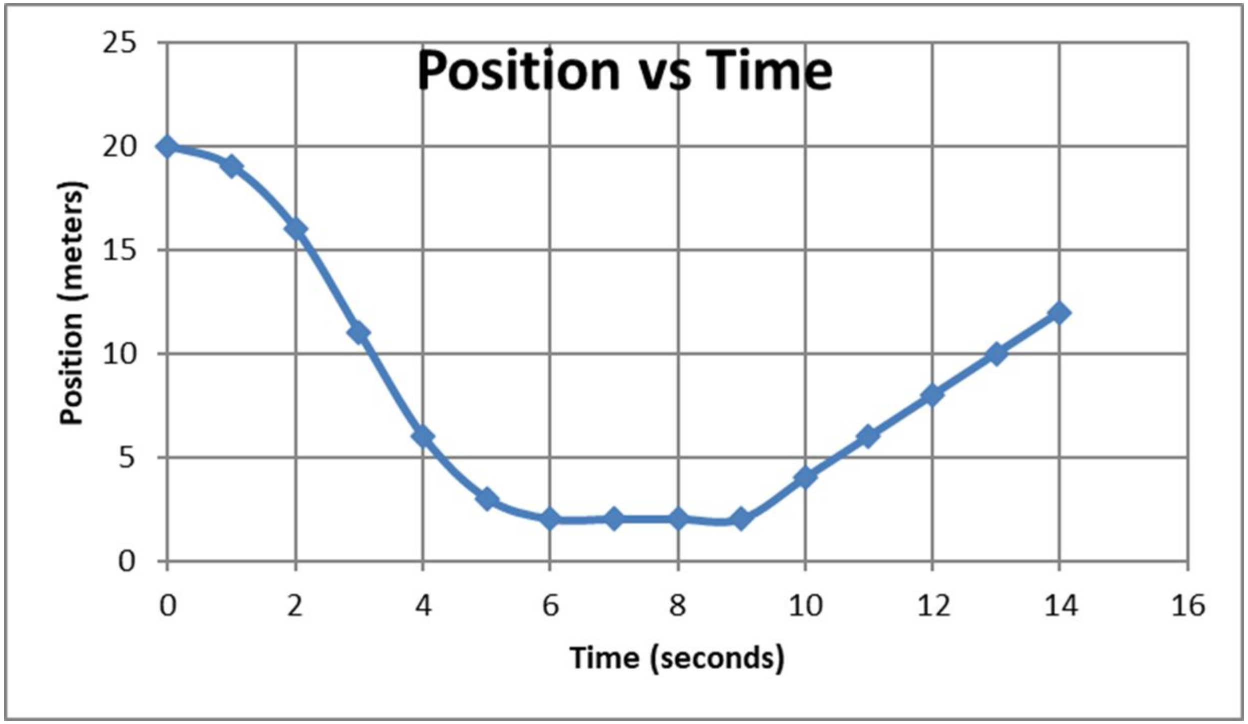

- Describe motion:

0-3 sec? Faster bwd

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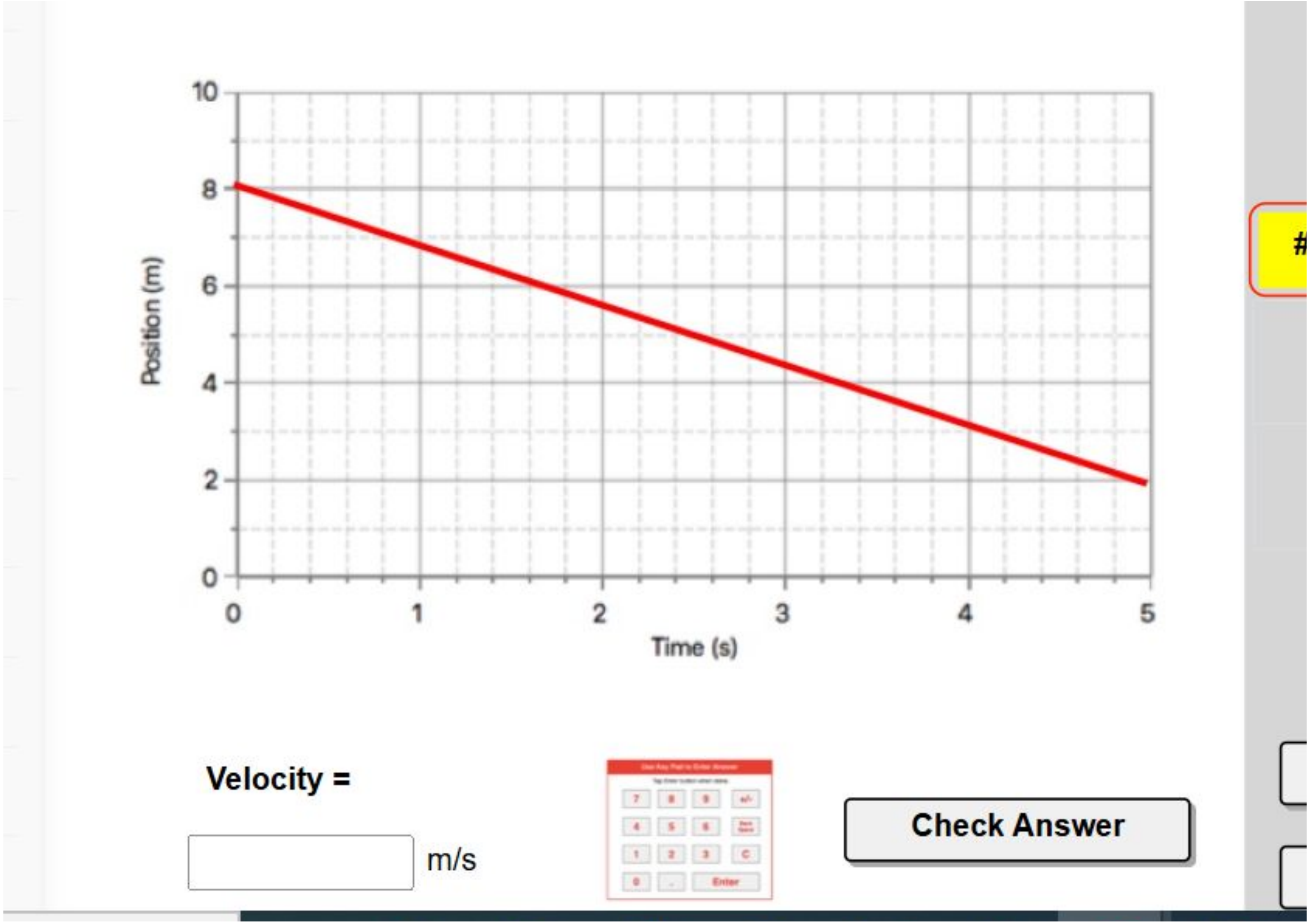
- 6-9 sec? At rest
- 9-14 sec? Con Vel fwd

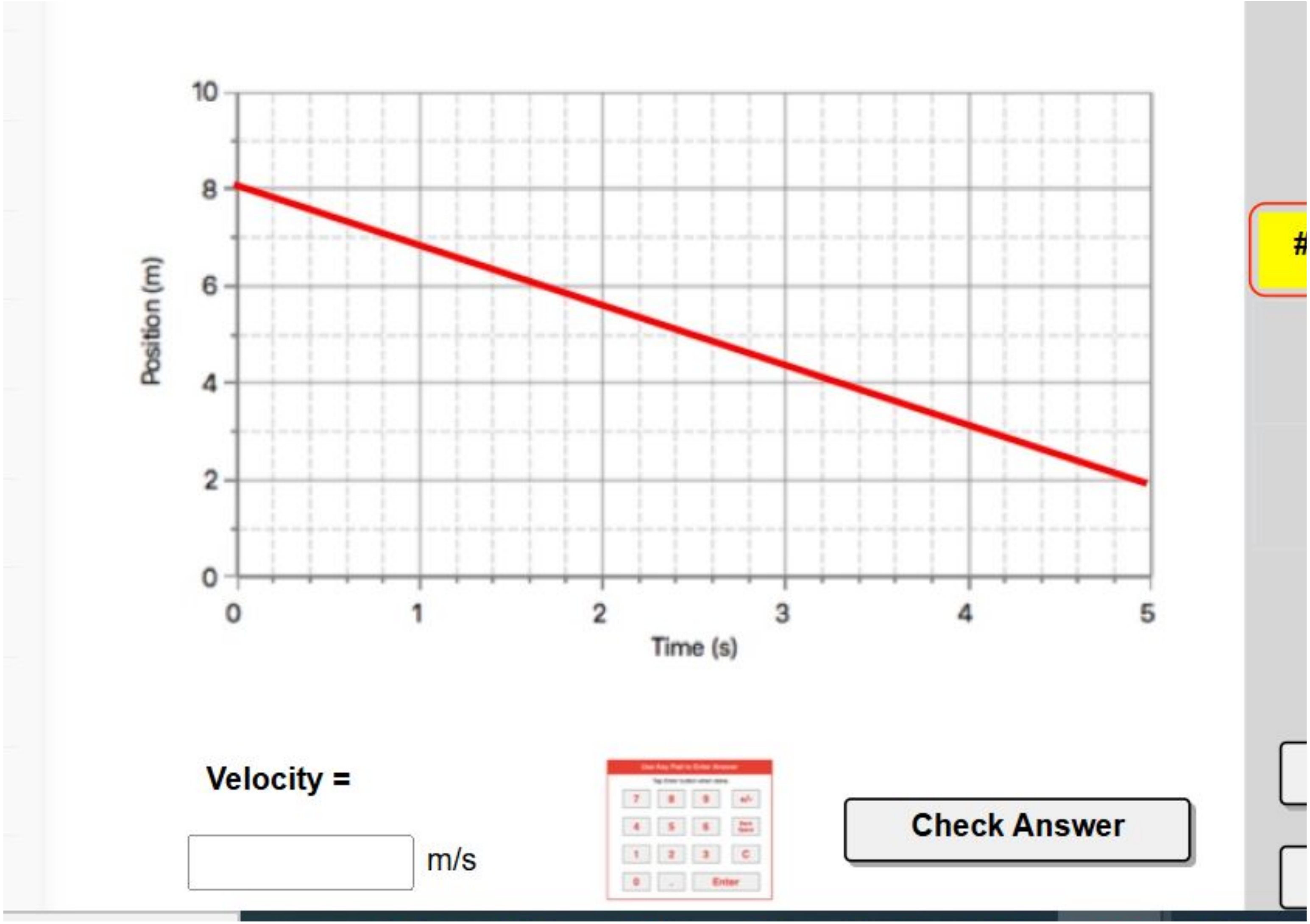
Find V (t = 9-14sec)

$V = (x-x_0)/t = (12-2)/5 = 2\ m/s$

What is the displacement? velocity (aka slope)?



displacement? 2-8 = -6 velocity (aka slope)? -6/5



Motion Graph Checkup

Turn in up front when finished

HW: Make sure you have completed all your Concept Builders from the week.

(Go to Task Tracker)

9/2 - 9/3 Agenda

Park phones

Position Graph Checkup feedback (makeups at 8 am & lunch)

Equations of motion

Lab Car & Ramp part 2

Check in with your neighbor. How was the holiday?

Learning Target

- I will be able to understand what types of motion are considered accelerating. I will utilize acceleration equations to analyze motion.

Section 3-1: Acceleration

- Acceleration is the rate at which velocity changes.
- In everyday language acceleration means “speeding up”
- In science acceleration refers to **speeding up,** *slowing down* or <u>changing directions</u>.
- Acceleration is a vector.

- Rocket sled with the Air Force’s Dr. Stapp (What can a human survive? Let’s find out!)
- <u>http://www.youtube.com/watch?v=3UEYxf4fl_A</u>

Acceleration

Acceleration – how fast you speed up, slow down, or change direction; it's the rate at which velocity changes. Two examples:

t (s)	*v* (mph)
0	55
1	57
2	59
3	61

t (s)	*v* (m/s)
0	34
1	31
2	28
3	25

$a = +2 \text{ mph/s}$

$a = -3 \, \frac{\text{m/s}}{\text{s}} = -3 \text{ m/s}^2$

Accelerated motion?

- Car goes from 0 to 50 MPH
- Car goes around a banked curve at a steady 50 MPH
- Car slows from 80 MPH to 55 MPH
- Car cruises at 55 MPH in a straight line

Useful Equations

- $V = d/t$ (for average velocity or constant velocity)
- $a = (v_f - v_i)/t$
- $v_f = v_i + at$
- $d = v_i t + \frac{1}{2} at^2$
- $v_f^2 = v_i^2 + 2ad$
- $d = \frac{1}{2}(v_i + v_f)t$

Key to symbols

d = displacement (meters, m)

t = time interval (seconds, s)

V_f = final velocity (m/s) – instantaneous

V_i = initial velocity (m/s) - instantaneous

a = acceleration (m/s/s) or the change in velocity (m/s^2)

Calculating acceleration

- To determine the acceleration of an object, you must calculate its change in velocity per unit of time
- Acceleration = $\frac{\text{Final velocity} - \text{initial velocity}}{\text{time interval}}$

- A roller coaster car starts down a slope, its velocity is 4 m/s but 3 second later its velocity is 25 m/s in the same direction, what is its acceleration?

$$\text{Acceleration} = \frac{\text{Final velocity} - \text{initial velocity}}{\text{time}}$$

Initial velocity = 4 m/s
Final velocity = 25 m/s
Time = 3 seconds

$$\text{Acceleration} = \frac{25 \text{ m/s} - 4 \text{ m/s}}{3 \text{ s}}$$

$a = 7 \text{ m/s}^2$

Burn-out! Find the acceleration of the truck!

- Goal: Calculate the acceleration (a) of the toy truck. Pull back (1-2 ft) and let it go! Solve for "a".
- Materials: meter stick, tape, stopwatch, toy truck.
- Useful tools:

- $d = v_i t + ½ \mathbf{a} t^2$

HW 9/2 - 9/3

- Finish the Acceleration & Velocity Practice Problem wkst. Bring to class on Thurs/Fri Show your work! Knowns, unknowns, substitute and solve equation….
- Finish any Concept Builders that remain in your Task Tracker (www.physicsclassroom.com)

9/4 - 9/5 Agenda

Park phones

Get a whiteboard & marker to share at your table

Set out your HW (Accel & Velo Prac Prob)

Freefall (Case study of acceleration)

Study Guide for Motion Test

Practice w/ symbols (declare each value as t, d, a, v_i or v_f)

1) A race car slows from 36 m/s to 15 m/s over 3.0 seconds. What is its average acceleration?

2) A car slows from 22 m/s to 3.0 m/s at a constant rate of 2.1 m/s/s. How many seconds are required before the car is traveling at 3.0 m/s?

3) A race car travels on a racetrack at 44 m/s and slows at a constant rate to a velocity of 22 m/s over 11 sec. How far does it move during this time?

Answers

1) A race car slows from 36 m/s to 15 m/s over 3.0 seconds. What is its average acceleration?

$a = (v_f - v_i)/t$

$a = (15 - 36)/3 = -7$ m/s/s

2) A car slows from 22 m/s to 3.0 m/s at a constant rate of 2.1 m/s/s. How many seconds are required before the car is traveling at 3.0 m/s?

$v_f = v_i + at$

$3 = 22 + (-2.1)t$ $t = 9$ sec

Answers

3) A race car travels on a racetrack at 44 m/s and slows at a constant rate to a velocity of 22 m/s over 11 sec. How far does it move during this time?

$d = ½ (v_i + v_f)t$

$d = ½ (22 + 44)(11)$

$d = 363$ m

Freefall (case study of accelerated motion)

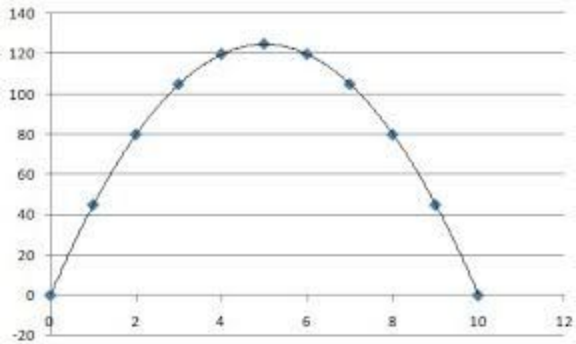
I jumped from space!

What does it mean to be in "freefall"?

Felix Baumgartner, professional sky-diver, goes for the world record

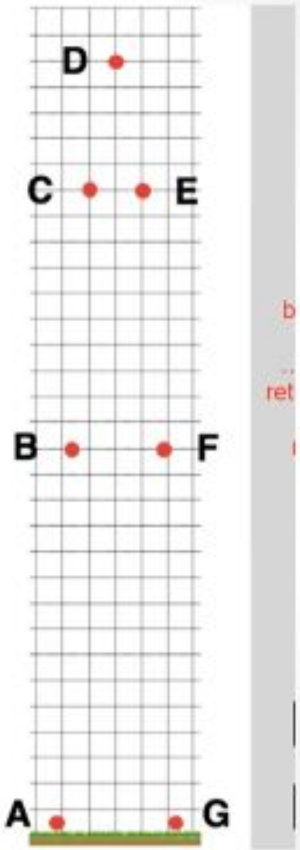
Freefall Notes (intro)

Demo book vs paper



Hammer vs Feather drop on the Moon

A ball is projected upward with an initial speed of approximately 30 m/s. The diagram at the right represents its position at 1-second intervals of time. At what location will the ball have a speed of 0 m/s?



At what location(s) will the ball have a speed of 10 m/s?

At what location(s) will the ball have a speed of 20 m/s?

At what location(s) will the ball have a speed of 30 m/s?

What is the acceleration at A,B,C,D?

A ball is projected upward with an initial speed of approximately 30 m/s. The diagram at the right represents its position at 1-second intervals of time. At what location will the ball have a speed of 0 m/s?

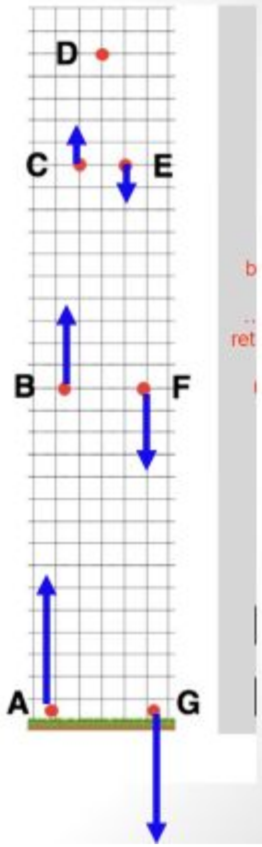
Location D

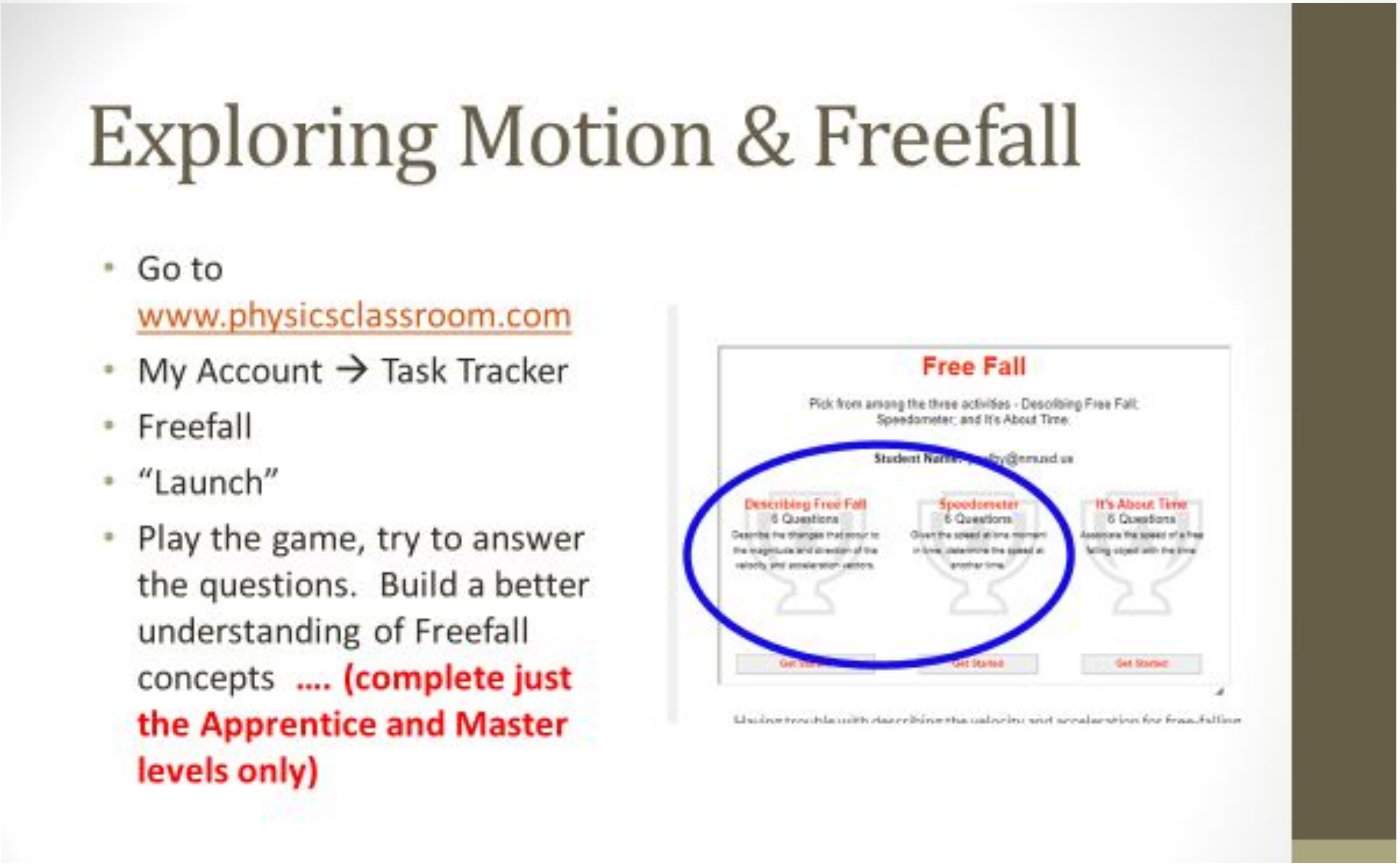
At what location(s) will the ball have a speed of 10 m/s? **C & E**

At what location(s) will the ball have a speed of 20 m/s? **B & F**

At what location(s) will the ball have a speed of 30 m/s? **A & G**

What is the acceleration at A,B,C,D? **Always 9.8 m/s/s downward (approx 10 m/s/s) Gravity never turns off!**





Check HW Accel & Velocity Prac Prob while they work…

Freefall practice problems

See worksheet

Link to Freefall Notes w solutions (check this out if you were absent)

HW:

1. Study Guide Motion (check answers posted on Schoology) Link to Study Guide solutions

2. Finish Concept Builders (check your Task Tracker)

3. Test Unit 0 on Monday **(bring a chromebook!)**

Stop by on Monday at Intervention if you have any questions…

9/8 Agenda

Review for Unit 0 Motion Test (grab a whiteboard to share)

Questions from Study Guide?

Return papers

HW: Finish all Concept Builders & study for the test (Tues/Wed)

Is Alcaraz the best ever?

9/9 - 9/10 Test Day!

Park phones

Chromebook ready (if no chromebook, then you will take paper version)

Calculator

Accel

- $v_f = v_i + at$
- $a = (v_f - v_i)/t$
- $d = v_it + ½\ at^2$

- $v_f^2 = v_i^2 + 2ad$
- $d = ½\ (v_i + v_f)t$

Freefall

$v_f = v_i + gt$

$d = v_it + ½\ gt^2$

$v = d/t$

$v_{avg} = (V_f - V_i)/2$

When finished, turn in your scratch paper and pick up a "Unit 1 Notes" packet

Top Fuel Dragsters Kings of Acceleration

- https://www.youtube.com/watch?v=-VF0JwxQqcA
- How do these machines work?
- Let’s take a look!

Useful Equations

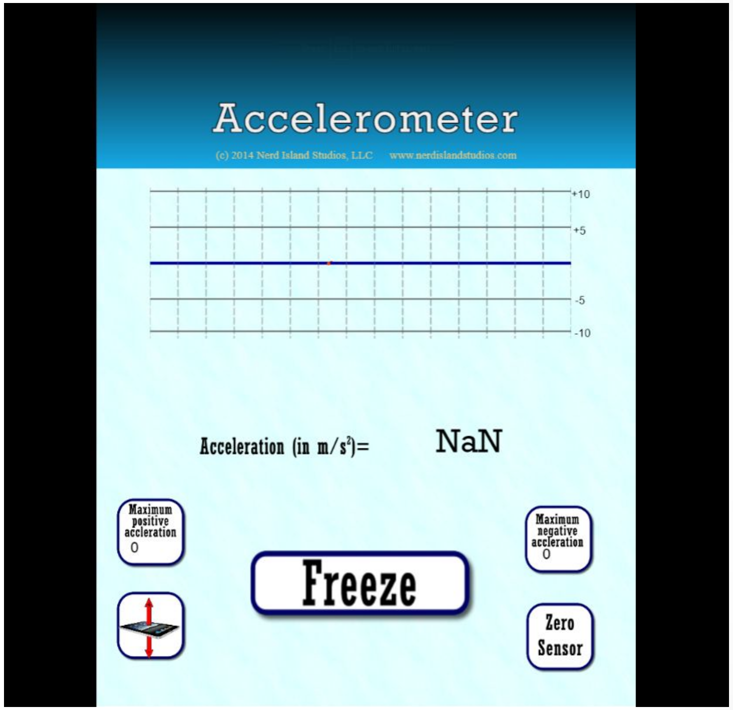
- $V = d/t$ (for average velocity or constant velocity)
- $a = (v_f - v_i)/t$
- $v_f = v_i + at$
- $d = v_i t + ½ at^2$
- $v_f^2 = v_i^2 + 2ad$
- $d = ½ (v_i + v_f)t$

Accel Notes: Predict the shape of graphs for a falling object

- We are about to conduct an experiment where we video record an object falling. This video app will then turn the motion into data for us to analyze. Make a prediction sketch of both the **Position vs. Time** and ***Velocity vs. Time*** graphs.

Feel the Acceleration!

- Using your phone (Chrome often works better)
- www.physicsclassroom.com
- Interactives
- 1-Dimension Kinematics
- Choose "Accelerometer"
- Launch!
- (Then explore the app, move and see what values come up. 9.8 m/s/s?)



Things to try with the app

- Can you create a freefall data set? (try a controlled drop in your hands)
- Does moving in a circle count as accel?
- What does this sensor do in your phone?
- How could we use this device to get useful insights?